

# Contextual Sparse Embedded Topic Modeling for Mass2Motif Discovery

## Abstract

Mass2Motifs are recurring fragment and neutral-loss patterns discovered by treating tandem mass spectra as short documents. Neural topic models infer spectrum-level topic mixtures efficiently, but this regime poses two competing requirements: each short spectrum should use few topics, while the collection should retain a broad motif inventory. We introduce Contextual Sparse ETM, an extension of the Embedded Topic Model, to address both requirements. The model retains ETM’s embedding-based topic–word construction, Gaussian latent variables and variational training framework, while modifying the decoder normalization and latent-to-simplex mapping. It uses fixed train-only spectral-word coordinates, balances fragment and neutral-loss decoder mass, adds contextual top-2 token-to-topic evidence to the posterior mean, and replaces dense document-topic softmax with 1.5-entmax. These adaptations add one learned scalar to a 19,278,000-parameter ETM.

We evaluated the model on a positive-ion-mode MS<sup>n</sup>Lib benchmark with compound- and scaffold-disjoint training, validation and test splits. On 7777 held-out test spectra, a representative 1000-topic run produced 731 motifs with associated molecules and a computable substructure overlap score (SOS), of which 424 had SOS at least 0.6. Tomotopy latent Dirichlet allocation (LDA) produced 605 evaluable and 365 useful motifs. Contextual Sparse ETM therefore increased these counts by 20.8% and 16.2%, respectively. The broader inventory had lower conditional mean SOS (0.6244 versus 0.6379) and a lower useful fraction among evaluable motifs (58.0% versus 60.3%). Completion negative log-likelihood, for which lower is better, was 9.5355 versus 9.7348. Across three training seeds on the same split, the model produced 731–740 evaluable and 415–432 useful motifs, with 3.68–3.74 effective topics per spectrum. The evidence therefore supports a reproducible gain in chemically assessable motif breadth, with explicit conditional-quality trade-offs rather than uniform superiority over LDA. A subsequent validation-only review supports one simpler ETM-based form: retain the conventional MLP Gaussian encoder, but replace leave-one-out context with the whole-spectrum mean. Across three paired seeds this gives 390.7 versus 389.7 useful motifs, with slightly better completion NLL and similarly compact mixtures. This recommendation is development evidence; the preceding held-out test results belong to the original leave-one-out model, not the newly simplified form. Global minimality is not established.

## 1 Introduction

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss features as words, and recurrent co-occurrence patterns as latent topics called Mass2Motifs [1]. A molecule may contain several reusable substructures, so a spectrum should be represented by a mixture of motifs rather than assigned to a single cluster. This probabilistic representation supports both exploratory substructure discovery and downstream chemical annotation [2].

Mass spectra are an unusually demanding document regime. A spectrum contains a small number of physical peaks, each peak can generate paired fragment and neutral-loss words, and normalized intensities are encoded as pseudo-counts. At the same time, large reference libraries motivate topic inventories with hundreds or thousands of components. A useful model must therefore meet three requirements at once: predict withheld spectral evidence, assign each short spectrum to only a few motifs, and retain many distinct topics across the corpus. Likelihood alone is insufficient because diffuse document mixtures or a small effective topic inventory can coexist with good prediction.

The Embedded Topic Model (ETM) is the base model used in this study. It retains an explicit topic–word distribution, represents topics and words in a shared embedding space, and replaces iterative document-level inference with a neural variational encoder [7]. Spectral-word embeddings are also well motivated in mass spectrometry: Spec2Vec, a learned mass-spectral similarity method, demonstrated that fragment and neutral-loss co-occurrence embeddings capture chemically useful relationships [22]. However, an ordinary ETM posterior does not explicitly use the local co-occurrence pattern surrounding each word in a short spectrum, and softmax cannot produce exact zero topic probabilities.

Contextual Sparse ETM extends that published ETM in three explicit steps. First, the decoder assigns equal total probability to fragment and neutral-loss channels so their unequal vocabulary sizes cannot set their

relative mass. Second, each observed word is combined with the leave-one-out mean of the other words in its spectrum, matched to the two closest ETM topic vectors, and aggregated into document-level evidence that adjusts the ETM posterior mean. Third, published 1.5-entmax replaces softmax to give exact document-topic zeros [21]. The multinomial mixture likelihood, Gaussian latent variational family and analytic standard-normal Kullback–Leibler (KL) divergence are retained. Entmax changes the distribution induced on topic mixtures; it is not merely a sparse estimator for an unchanged logistic-normal ETM. Channel balancing and entmax add no parameters; contextual evidence adds one learned scalar. Controlled synthetic component comparisons test context and entmax separately, while the real-data canonical-versus-balanced ETM comparison isolates channel balancing.

This study makes three contributions:

1. a fully specified Contextual Sparse ETM that preserves explicit, inspectable Mass2Motif topic–word distributions;
2. a leakage-controlled benchmark covering data acquisition, spectral preprocessing, train-only token embeddings, document completion, MAG and SOS; and
3. truth-known synthetic validation together with three-seed real MS<sup>n</sup>Lib test evaluation of predictive fit, chemical utility, sparsity, global topic use and computational load.

## 2 Related work

**Probabilistic and neural topic models.** Latent Dirichlet allocation (LDA) models each document as a mixture of categorical topic–word distributions [4]. Neural Variational Document Models and Autoencoding Variational Inference for Topic Models (AVITM) established amortized variational inference, in which one encoder replaces separate iterative inference for every document. They also showed the practical importance of maintaining useful latent components [5, 6]. ETM subsequently parameterized topic–word scores through topic and word embedding inner products while retaining a probabilistic document model [7]. The present method adapts this embedding decoder and variational structure.

ProdLDA is another compact published neural option, but it mixes topic scores in natural-parameter space, giving a product of experts rather than ETM’s additive mixture [6]. Embedding Clustering Regularization Topic Model (ECRTM) directly addresses topic collapse with an additional embedding-clustering objective [16]. Both are legitimate external baselines; neither can be represented faithfully by simply renaming a small ETM encoder modification. Direct-Dirichlet variational autoencoders offer a different neural route to topic mixtures without a Gaussian approximation or entmax [11]. Our expanded review tests ProdLDA, direct-Dirichlet models and additive LDA/ETM adaptations, separating published architecture from implementation and benchmark-specific choices.

**Short-document topic inference.** Short-text models emphasize that individual documents provide limited co-occurrence evidence. The Biterm Topic Model aggregates word-pair statistics at corpus level [17], while the Neural Sinkhorn Topic Model uses embedding geometry and optimal transport, a geometry-aware distribution-matching objective, to improve neural topic modeling of short documents [18]. Empirical studies further show that document length changes the relationship between topic-model hyperparameters and evaluation measures [19]. Contextual Sparse ETM addresses the same statistical regime while preserving per-spectrum multi-topic mixtures and the additive categorical-mixture observation model.

**Sparse probability mappings.** Sparsemax and the entmax family replace dense softmax probabilities with probability-vector mappings that can return exact zeros [20, 21]. 1.5-entmax lies between softmax and sparsemax and remains differentiable wherever its probabilities are non-zero. In the proposed model it acts only on document–topic logits; the topic–word decoder remains probabilistic and dense within each spectral channel. Gaussian-to-sparse-simplex topic models have direct published precedent: NSMDM/NSMTM use Gaussian–sparsemax constructions and relaxed-Wasserstein regularization [13]. Our entmax construction is not a reproduction of that objective: the ELBO is evaluated in the nonsingular Gaussian latent space, where its KL is well defined, while the additive embedding decoder retains full vocabulary support within each channel.

**Attention and explicit selection alternatives.** TAN-NTM uses topic-aware attention over recurrent token representations for variational inference [14]; topic-informed recognition is therefore not a new general principle. The present spectral encoder uses order-invariant weighted word/context directions instead of a recurrent sequence encoder. SpareNTM adds Bernoulli topic selectors and non-mean-field Dirichlet inference, with a continuous sampling relaxation [15]. This is a published alternative for sparse mixtures, but introduces a selector prior and another latent family. These architectures provide relevant precedent and future comparators; they are reviewed here, not newly fitted, and no MS<sup>n</sup>Lib performance claim is made for them. Their existence is a further reason not to equate this bounded deletion study with global minimality.

**Mass-spectrometry representation and evaluation.** MS2LDA introduced the spectrum-as-document formulation [1]. MS2LDA 2.0 scaled the workflow through Tomotopy and introduced Mass2Motif Annotation Guidance (MAG) and the substructure overlap score (SOS) for chemical evaluation [2]. MS<sup>n</sup>Lib provides the open reference-standard spectra used here [3], and Spec2Vec supplies the downstream retrieval representation used by MAG [22]. These established components define the domain task and its chemical evidence surface.

Together, these lines of work define the design target for this study: retain ETM’s explicit topic–word generator and the established MS2LDA chemical evaluation workflow, while adapting document inference to the limited co-occurrence evidence in short spectra. The proposed model changes the ETM posterior, latent-to-simplex link and channel normalization to meet that target without introducing a separate clustering objective.

### 3 Materials and methods

#### 3.1 MS<sup>n</sup>Lib source, acquisition and loading

MS<sup>n</sup>Lib is an open reference-standard library generated from seven compound collections using automated multi-stage fragmentation acquisition and processing. The complete resource contains more than 2.3 million MS<sup>n</sup> spectra, including more than 357,000 MS<sup>2</sup> spectra for 30,008 unique structures across both ionization modes [3]. We used the positive-ion-mode benchmark input distributed with the MS2LDA 2.0 paper rather than resampling the complete resource [2].

The frozen assets are distributed in Zenodo record 20179680 [27]. The acquisition script downloads `Data.zip` and `model_positive_mode.zip`, verifies the declared byte size and MD5 checksum, checks ZIP-archive integrity and paths, extracts through a staging directory, verifies required files, and writes an acquisition manifest. The topic-model input is `Data/Benchmark_MSn_Lib/Corinna_Library_filtered_positive.mgf`. The positive-ion-mode Spec2Vec model, embedding matrix and SQLite spectral-library database are used only for downstream MAG evaluation. Known structures therefore support split construction and chemical scoring but are not labels for topic-model fitting.

The streaming Mascot Generic Format (MGF) reader treats each `BEGIN IONS/END IONS` block as one spectrum, parses peak lists rather than trusting `NUM_PEAKS`, and requires a Universal Spectrum Identifier, feature identifier, SMILES molecular-structure string, InChIKey structure identifier and precursor mass-to-charge ratio ( $m/z$ ). The RDKit cheminformatics toolkit reparses each SMILES string and verifies agreement between the derived connectivity InChIKey and the deposited identifier. Figure 1 summarizes the complete hand-off from public data to model matrices.

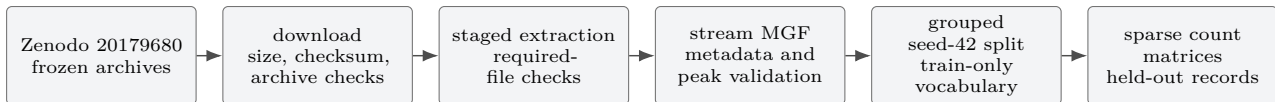


Figure 1: Reproducible MS<sup>n</sup>Lib ingestion. The Mascot Generic Format (MGF) file supplies spectral documents; verified Spec2Vec similarity-retrieval assets branch from the extraction only for leakage-filtered chemical annotation.

#### 3.2 Dataset construction and leakage control

The input MGF contains 41568 positive-ion-mode spectra. Within each spectrum, finite peak intensities are divided by the largest peak intensity. Peaks outside  $m/z \in (0, 2000]$  or normalized intensity  $[0.01, 1]$  are removed. If more than 1,000 peaks remain, the most intense are retained, and spectra with fewer than five retained peaks are discarded. This deterministic procedure retains 38888 spectra.

Every physical peak generates a fragment word and, when the precursor-minus-peak mass exceeds 0.01 Da, a neutral-loss word. Both masses are rounded to two decimal places, producing tokens such as `frag@70.04` and `loss@162.05`, consistent with MS2LDA 2.0 [2]. A peak with normalized intensity  $I$  contributes  $\text{round}(100I)$  copies of each applicable word. The resulting document is a count-valued bag of spectral words. Fragment and neutral-loss tokens from the same peak are additionally retained as one physical peak group for document-completion splitting.

To prevent structurally related compounds from crossing partitions, molecules are keyed by the connectivity block of the InChIKey. Cyclic molecules are grouped by their non-chiral Bemis–Mureko core ring-and-linker scaffold [24], while each acyclic connectivity group remains intact. The 28572 resulting groups are sorted by descending size and assigned deterministically toward 70/10/20 train, validation and test targets using seed 42. This produces 27222 training, 3889 validation and 7777 test spectra from 38465 connectivity groups, with zero compound or split-group overlap.

The vocabulary is built from training spectra only, requires each token to occur in at least three training spectra (document frequency at least three), preserves first-seen training order, and contains 21233 fragment

and neutral-loss tokens. Every split is stored in the same fixed column order as a SciPy compressed sparse-row (CSR) count matrix. All learned parameters, token embeddings, vocabulary entries and co-occurrence statistics use only the training partition. Model formulation, hyperparameters and component comparisons use training and validation evidence only. The test matrices are used after these choices and fitted checkpoints are fixed; test evaluation performs no fitting, model selection or spectrum-specific optimization. The subsequent simplification review explicitly uses validation only after these historical test results were available. It does not make that test split newly unseen or transfer the frozen model’s test scores to a different architecture.

For predictive evaluation, validation and test spectra are each deterministically divided 50/50 by whole physical peak groups. A fragment and its corresponding neutral loss therefore cannot appear on opposite sides. The observed half is renormalized without consulting the maximum intensity of the withheld half. Out-of-vocabulary withheld words are recorded but excluded from the likelihood numerator and denominator. The completion result contains 7776 eligible spectra and 2,534,703 in-vocabulary withheld token copies; the withheld out-of-vocabulary fraction is 2.69%.

Finally, each vocabulary token has a fixed 48-dimensional skip-gram with negative sampling (SGNS) coordinate trained only on the training documents [23]. For five epochs, eight count-weighted positive pairs are sampled per document. Every positive pair uses five negative words drawn with probability proportional to corpus frequency raised to 0.75. Source embeddings are initialized uniformly on  $[-0.5/48, 0.5/48]$ , context embeddings at zero, and the PyTorch SparseAdam optimizer uses learning rate 0.01 with batches of 8,192 pairs. Source and context embeddings are averaged and normalized after the fifth epoch. Molecular identifiers and structures do not enter this representation. Holding these coordinates fixed separates the geometry supplied to every ETM variant from the learned topic vectors and posterior, so component comparisons cannot be attributed to variant-specific word-embedding updates.

### 3.3 Contextual Sparse ETM

#### 3.3.1 Model lineage and design rationale

Contextual Sparse ETM begins with the published Embedded Topic Model of Dieng et al. [7]. ETM supplies the generative and variational starting point: topic and word vectors share an embedding space, their inner products define an explicit topic–word distribution, a neural encoder parameterizes a Gaussian document posterior, and a multinomial likelihood with analytic Gaussian KL divergence trains the model end to end. We retain these components, change decoder normalization, and replace the softmax link from Gaussian latents to topic mixtures. The resulting model remains ETM-based but is not the identical published ETM generative distribution.

The adaptations in Table 1 address properties specific to short tandem mass spectra. Fixed train-only SGNS coordinates provide the spectral-word geometry consumed by ETM. Channel balancing prevents unequal fragment and neutral-loss vocabulary sizes from setting their total decoder mass. Contextual evidence lets the other peaks in one short spectrum disambiguate each token. Finally, published 1.5-entmax supplies exact per-spectrum sparsity. Only the context scalar is learned in addition to the ETM parameters.

Table 1: Lineage of Contextual Sparse ETM. “Inherited” means retained from the published Embedded Topic Model (ETM). The canonical-versus-balanced real comparison isolates channel balancing; the synthetic component study isolates contextual evidence and entmax.

| Component                                                              | Provenance                   | Purpose                                                                                                |
|------------------------------------------------------------------------|------------------------------|--------------------------------------------------------------------------------------------------------|
| Embedding scores, encoder, Gaussian latents, mixture likelihood and KL | Inherited [7]                | Preserve a published probabilistic topic model with explicit $\beta$ and amortized document inference. |
| Train-only skip-gram (SGNS) word coordinates                           | ETM input configuration [23] | Supply leakage-free fragment/loss co-occurrence geometry.                                              |
| Fragment/loss-balanced $\beta$                                         | Added; no parameters         | Give the paired spectral channels equal aggregate probability mass.                                    |
| Contextual top-2 posterior evidence                                    | Added; one scalar            | Use neighboring peaks to disambiguate tokens in a short spectrum.                                      |
| 1.5-entmax document mixture                                            | Published mapping [21]       | Change the latent-to-simplex link to permit exact document-topic zeros.                                |

#### 3.3.2 Generative specification and channel-balanced ETM decoder

Let  $d \in \{1, \dots, D\}$  index spectra,  $w \in \{1, \dots, V\}$  index words and  $k \in \{1, \dots, K\}$  index topics. Let  $c_{dw} \geq 0$  be the raw spectral-word pseudo-count and

$$x_{dw} = \frac{c_{dw}}{\sum_v c_{dv}} \quad (1)$$

the normalized encoder input. Each word has a fixed SGNS vector  $\rho_w \in \mathbb{R}^{48}$ , and each topic has a learned ETM vector  $\alpha_k \in \mathbb{R}^{48}$ . The stored SGNS rows are unit-normalized before model construction. Throughout,

$$\text{normalize}(v) = \frac{v}{\max(\|v\|_2, \epsilon)}, \quad \text{center}(v) = v - \frac{1}{K} \mathbf{1} \mathbf{1}^\top v, \quad \epsilon = 10^{-12}.$$

The topic-word distribution follows the ETM inner-product construction [7]. Fragment and neutral-loss channels are normalized separately so vocabulary size cannot determine their aggregate mass. With  $V_F$  and  $V_L$  the two channel vocabularies,

$$\beta_{kw} = \begin{cases} \frac{1}{2} \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v \in V_F} \exp(\alpha_k^\top \rho_v)}, & w \in V_F, \\ \frac{1}{2} \frac{\exp(\alpha_k^\top \rho_w)}{\sum_{v \in V_L} \exp(\alpha_k^\top \rho_v)}, & w \in V_L. \end{cases} \quad (2)$$

Thus  $\sum_w \beta_{kw} = 1$ , each topic assigns exactly one half of its probability to fragments and one half to neutral losses, and token rankings within each channel remain learned. This channel constraint adds no parameters.

For a spectrum encoded as  $N_d = \sum_w c_{dw}$  integer pseudo-counts, the generative model is

$$z_d \sim \mathcal{N}(0, I), \quad \theta_d = \text{entmax}_{1.5}(z_d), \quad (3)$$

$$c_d \mid z_d, N_d, \beta \sim \text{Multinomial}(N_d, \beta^\top \theta_d). \quad (4)$$

Equivalently, each encoded token selects a topic from  $\theta_d$  and a word from that topic's  $\beta_k$ . Published ETM uses  $\text{softmax}(z_d)$  at the corresponding step. Entmax induces a different prior on the simplex, including boundary mass at exact zeros;  $\theta_d$  is therefore not logistic-normal. Contextual evidence below modifies only the variational posterior, not the generative prior.

The ordinary ETM encoder is a two-layer rectified network,

$$e_d = \text{ReLU}(W_2 \text{ReLU}(W_1 x_d + b_1) + b_2), \quad (5)$$

$$\mu_d = W_\mu e_d + b_\mu, \quad \log \sigma_d^2 = W_\sigma e_d + b_\sigma, \quad (6)$$

with two hidden layers of width 800.

### 3.3.3 Contextual evidence, sparse mixtures and variational training

Short spectra provide most of their disambiguating information through the other words observed alongside a token. Define normalized word and topic vectors  $\hat{\rho}_w = \text{normalize}(\rho_w)$  and  $\hat{\alpha}_k = \text{normalize}(\alpha_k)$ . For every non-zero word in spectrum  $d$ , its weighted leave-one-out context is

$$\bar{\rho}_{d \setminus w} = \frac{\sum_v x_{dv} \hat{\rho}_v - x_{dw} \hat{\rho}_w}{\max(1 - x_{dw}, \epsilon)}. \quad (7)$$

The contextual word representation is

$$h_{dw} = \text{normalize}(\hat{\rho}_w + c \bar{\rho}_{d \setminus w}), \quad (8)$$

where  $c$  is one learned scalar.

The word is scored against the same topic vectors that define the decoder:

$$a_{dwk} = h_{dw}^\top \hat{\alpha}_k / \tau_c, \quad \tau_c = 1.0. \quad (9)$$

Let  $\mathcal{T}_{dw}$  contain the two largest scores. The local contextual assignment is

$$q_{dwk} = \begin{cases} \frac{\exp(a_{dwk})}{\sum_{j \in \mathcal{T}_{dw}} \exp(a_{dwj})}, & k \in \mathcal{T}_{dw}, \\ 0, & \text{otherwise.} \end{cases} \quad (10)$$

Count-weighted aggregation yields one document evidence distribution,

$$r_{dk} = \sum_w x_{dw} q_{dwk}. \quad (11)$$

Because every local assignment and each  $x_d$  are normalized,  $r_d$  lies on the topic probability simplex: its entries are non-negative and sum to one. The implementation renormalizes its row sum once more to remove finite-precision accumulation error.

The document evidence is converted to a bounded, row-centered posterior offset:

$$o_d = \text{center}[\log(r_d + \frac{1}{K})], \quad \tilde{\mu}_d = \mu_d + o_d. \quad (12)$$

The fixed  $1/K$  pseudocount bounds evidence for topics outside the local top-2 assignments, while centring makes uniform evidence an exact no-op. The variational sample and document mixture are

$$z_d = \tilde{\mu}_d + \exp(\frac{1}{2} \log \sigma_d^2) \odot \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I), \quad (13)$$

$$\theta_d = \text{entmax}_{1.5}(z_d). \quad (14)$$

1.5-entmax supplies exact zeros while preserving a normalized mixture. More explicitly, it computes

$$\theta_{dk} = \left[ \frac{z_{dk}}{2} - \tau_d \right]_+^2, \quad \sum_k \left[ \frac{z_{dk}}{2} - \tau_d \right]_+^2 = 1, \quad (15)$$

where the threshold  $\tau_d$  is determined from the sorted logits. The code then divides by the finite-precision row sum, which preserves all exact zeros. At deterministic inference,  $z_d = \tilde{\mu}_d$ . The resulting  $\hat{\theta}_d = \text{entmax}_{1.5}(\tilde{\mu}_d)$  is a plug-in estimate, generally different from the posterior expectation  $\mathbb{E}_q[\text{entmax}_{1.5}(z_d)]$ .

The conditional word probability remains the usual topic mixture,

$$p(w | d) = \sum_{k=1}^K \theta_{dk} \beta_{kw}. \quad (16)$$

Write  $q_d(z) = \mathcal{N}(\tilde{\mu}_d, \text{diag}(\sigma_d^2))$ . Apart from the count-dependent multinomial constant, the evidence lower bound (ELBO) is

$$\text{ELBO}_d = \mathbb{E}_{q_d} \left[ \sum_w c_{dw} \log \left( \sum_k \text{entmax}_{1.5}(z_d)_k \beta_{kw} \right) \right] - \text{KL}[q_d \| \mathcal{N}(0, I)]. \quad (17)$$

This bound is formulated in Gaussian latent space. It does not require an invertible entmax transformation, a simplex density, or a change-of-variables Jacobian. Sharing topic vectors between the encoder correction and decoder restricts the parameterization of  $q_d$  but does not invalidate the bound.

The stochastic loss supplied to Adam is the raw pseudo-count multinomial reconstruction (up to its parameter-independent combinatorial constant) plus the analytic standard-normal Gaussian KL divergence. One reparameterized Gaussian draw is used per document and minibatch update:

$$\mathcal{L}_{\text{rec}} = -\frac{1}{D} \sum_d \sum_w c_{dw} \log p(w | d), \quad (18)$$

$$\mathcal{L}_{\text{KL}} = -\frac{1}{2D} \sum_d \sum_k (1 + \log \sigma_{dk}^2 - \tilde{\mu}_{dk}^2 - \exp(\log \sigma_{dk}^2)), \quad (19)$$

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \mathcal{L}_{\text{KL}}. \quad (20)$$

Adam additionally applies coupled weight decay  $\lambda = 1.2 \times 10^{-6}$  to all learned parameters, i.e. it adds  $\lambda\phi$  to the gradient for parameter vector  $\phi$ . No auxiliary co-occurrence, topic-separation or assignment loss is used.

Figure 2 shows how contextual evidence adjusts the Gaussian posterior while the embedding decoder retains an explicit topic-word output.

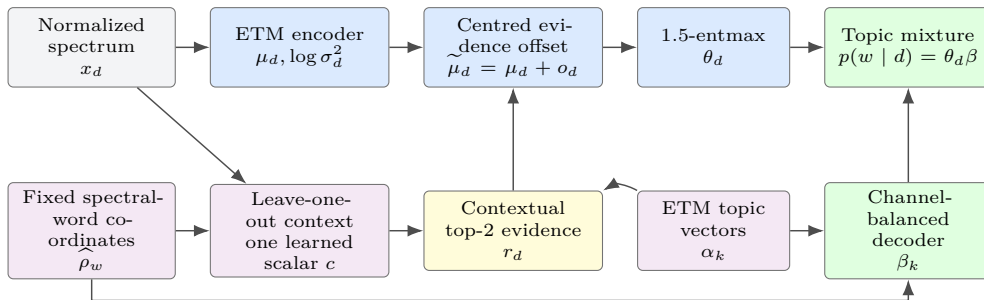


Figure 2: Contextual Sparse Embedded Topic Model (ETM). Contextual token assignments use the same fixed word and learned topic coordinates as the ETM topic-word decoder, then adjust the Gaussian document posterior before 1.5-entmax produces a sparse topic-probability vector. Solid arrows denote computations used at both training and inference. Context scoring normalizes topic vectors; the decoder uses their full vectors. Training draws a Gaussian latent, whereas evaluation uses the posterior-mean plug-in.



For auditability, the executable implementation follows this notation as a sequence of named tensor functions: channel-balanced topic-word probabilities, leave-one-out context, contextual top-2 evidence, the centered log-evidence offset, diagonal-Gaussian KL divergence, 1.5-entmax and raw-count reconstruction each have one direct implementation listed in Table 17. A single thin PyTorch module registers the learned matrices, vectors and scalar needed for checkpointing; it contains no selectable model variants. Deterministic inference calls the same posterior sequence as training but sets  $z_d = \tilde{\mu}_d$  rather than drawing  $\varepsilon$ .

With vocabulary size  $V$ , hidden width  $H$ , topics  $K$  and embedding dimension  $L$ , the learned-parameter count is

$$VH + H + H^2 + H + 2(KH + K) + KL + 1.$$

For  $V = 21,233$ ,  $H = 800$ ,  $K = 1000$  and  $L = 48$ , the model has 19,278,001 learned parameters, compared with 19,278,000 for the channel-balanced ETM; fixed  $\rho$  rows are not counted. The only addition is  $c$ , which converged to 0.2089 in the representative run. Models are trained for 120 fixed epochs with Adam, learning rate 0.005, weight decay  $1.2 \times 10^{-6}$  and batch size 256. PyTorch deterministic algorithms are enabled, non-finite objectives or gradients terminate execution, and six host worker threads feed training batches to an NVIDIA CUDA-compatible graphics processing unit (GPU).

At validation and final test evaluation, the observed half-spectrum is encoded deterministically for document completion, while the full spectrum is encoded for topic-inventory and chemical analyses. All 1000 topic-word distributions are exported for MAG. Training and inference use the same equations; no spectrum-specific optimization loop is required. Completion NLL evaluates the word probabilities from the deterministic plug-in mixture inferred from the observed half. It is not an integrated posterior-predictive likelihood or an estimate of the model’s exact marginal evidence.

### 3.4 Experimental design and comparators

The simulator contains 18 planted motifs with paired fragment and neutral-loss words. Each spectrum contains one to three generating motifs. Training-only vocabulary construction and SGNS match the real workflow, and intensities use the same raw pseudo-count reconstruction. The primary experiment uses 800 training and 160 validation spectra, fits  $K = 36$  topics, and averages seeds 11, 23 and 37. A high-overcompleteness experiment, in which fitted topics greatly outnumber planted motifs, fixes seed 11 and increases the fitted inventory to  $K = 128$  while retaining 18 planted motifs.

The synthetic endpoints are completion NLL, cosine recovery of the planted topic-word matrix, cosine recovery of the planted document mixtures, dominant motif accuracy, median effective topics, materially active topics and unique top-1 topics. Dominant motif accuracy compares the largest fitted mixture component with the largest planted component; a top-1 topic is the fitted topic with the largest weight in one spectrum. Optimal one-to-one matching aligns fitted topics to planted motifs before recovery is calculated.

For the real-data experiment, four methods are fitted to the same training matrix, compared on the same validation matrices, frozen and finally evaluated on the same test matrices:

1. canonical ETM: fixed SGNS coordinates, global topic-word softmax and dense document-topic softmax;
2. fragment/loss-balanced ETM: the same model with separate 50/50 fragment and neutral-loss decoder normalization;
3. Contextual Sparse ETM: the balanced decoder plus contextual top-2 posterior evidence and 1.5-entmax document mixtures; and
4. Tomotopy 0.13.0 LDA: the established MS2LDA-style comparator [26].

The first three rows therefore follow the model lineage from published ETM to the channel-balanced decoder and then the complete contextual sparse posterior. The two ETM baselines use the same 48-dimensional embeddings, two-layer 800-unit encoder,  $K = 1000$ , raw pseudo-count objective, Adam settings, batch size, 120 epochs and training seed 7043. Tomotopy uses  $K = 1000$ ,  $\alpha = 0.6$ ,  $\eta = 0.1$  and one worker in its deterministic no-parallel mode (`ParallelScheme.NONE`). It trains until convergence or 5000 iterations, ending at 750 iterations in this run; validation and test inference use 100 iterations under the same deterministic no-parallel setting.

Initialization robustness is assessed by repeating the proposed model with training seeds 7043, 23 and 37. The data split, vocabulary, SGNS embeddings, configuration, epoch count and chemical evaluation remain fixed. These repeats measure sensitivity to model initialization and minibatch order, not uncertainty over data splits.

### 3.5 Validation-only model-simplification review

A subsequent review asks whether fewer mechanisms can retain chemically useful breadth and compact neural topic mixtures. It preserves the historical test comparison and does not evaluate new models on test spectra. The review first audits an earlier ETM development screen using posterior-head batch normalization (BN) and the AVITM diagonal Laplace approximation to a symmetric Dirichlet prior [6]. For per-topic concentration  $a = 0.02$ , the approximate Gaussian prior has mean zero and variance  $(1 - 1/K)/a$ . This is not an exact Dirichlet prior, does not create exact zeros, and has total Dirichlet concentration  $Ka$ , which changes with  $K$ .

Two further simplifications are then evaluated. *BN-entmax ETM* retains the single-vocabulary ETM softmax decoder and standard-normal latent prior, normalizes the encoder’s mean and log-variance heads, and uses the entmax link in Equation 3. It has no contextual evidence, channel constraint or auxiliary loss. Both learned and fixed-unit normalization scales are screened; biases remain trainable. Final normalization moments are recomputed on the entire training matrix with model weights fixed, then frozen for evaluation. No validation rows set those moments. This inference-state requirement is part of the recipe, not free architectural simplicity. *Contextual ETM without channel balancing* changes only Equation 2 to a single softmax over all words, retaining context, entmax, the prior and the training objective. These are explicit ETM adaptations, not reproductions of the complete AVITM or ProdLDA models.

Before the new runs, a bounded protocol fixes 120 epochs, hidden width 800, batch size 200, raw counts, learning rate 0.005, Adam first-moment coefficient 0.9, and weight decay  $1.2 \times 10^{-6}$ . Synthetic inputs and train-only SGNS coordinates are reused. Candidates first pass seed 11 at  $K = 36$  and 128, then repeat seeds 23 and 37 at both  $K$  values with paired current-model runs. The three-seed mean must retain beta recovery within 0.05 and theta recovery within 0.10 of the current model, with median effective topics at most five and no catastrophic duplicate run. At  $K = 128$  it must retain at least 90% of recovered motifs (matched beta cosine at least 0.50, out of 18 planted). These are development tolerances, not non-inferiority confidence bounds.

At most two distinct simplifications advance to the same 3,889-spectrum MS<sup>n</sup>Lib validation set at  $K = 1000$ , initialization seed 7043. A replacement must retain at least 95% of both useful and evaluable counts, lose at most 0.02 mean SOS, worsen completion NLL by at most 2%, retain at least 800 winning topics, and have median effective topics at most five without catastrophic duplication. Because the historical real runs used batch size 256, the review adds a paired current-model run at 200 before inspecting new real results. Retention must hold against both references. A passing candidate would still require two more real training seeds before replacement. The strict duplicate flag uses a component covering at least half of all topics at cosine at least 0.999; it is not evidence that every topic is distinct.

**Expanded published-family screen.** A separately recorded second phase broadens the search after the initial BN-entmax result. The official Pyro ProdLDA tutorial supplies a compact reference architecture [12]: two 100-unit softplus encoder layers, Gaussian mean/log-variance heads, standard-normal prior, affine-free BN, 0.2 dropout, and a product-of-experts (PoE) decoder. We port its tensor computation to PyTorch and optimize analytic Gaussian KL plus one reconstruction sample; we do not run Pyro stochastic variational inference (SVI) or reproduce its text-corpus scores.

The direct-Dirichlet family uses one 100-unit ReLU layer, 0.25 dropout, a unit-scale/learned-bias BN concentration head, softplus with floor  $10^{-5}$ , per-topic prior  $a_0 = 0.02$ , implicit-reparameterized Dirichlet samples and analytic KL [11]. Its three decoder variants are a BN PoE decoder, a free additive LDA topic matrix, and ETM’s fixed-SGNS embedding decoder. The latter two are explicit adaptations, not the paper’s stock model. For an additive variant,

$$q_\phi(\theta_d | c_d) = \text{Dir}(a_\phi(c_d)), \quad p(\theta_d) = \text{Dir}(a_0 \mathbf{1}), \quad p(w = v | \theta_d) = \sum_k \theta_{dk} \beta_{kv}. \quad (21)$$

The variational mean is exactly  $a_\phi / \sum_k a_{\phi k}$ ; a sparse prior does not guarantee that this inferred mean is compact.

These four recipes use raw counts, batch 200, 120 epochs, Adam learning rate 0.001/first-moment coefficient 0.9 and no weight decay. The Dirichlet variants have 100-epoch linear KL warmup. Training exponential-moving-average (EMA) statistics remain frozen at inference, without the first phase’s recalibration. All use synthetic seeds 11, 23 and 37 at both  $K$  values. The neural widths and optimization recipes deliberately follow their cited families, so this is not an equal-capacity or exhaustively tuned ranking. Because synthetic truth is additive, ProdLDA receives a real-validation baseline run even if it fails the synthetic replacement gate; at most one passing Dirichlet candidate can also advance. The initial real replacement criteria are unchanged.

After inspecting this complete screen, a single follow-on hypothesis is recorded before training: replace only ProdLDA’s latent softmax by 1.5-entmax, retaining its other components and recipe. This is a new published-component combination, not stock ProdLDA. Its Gaussian latent ELBO remains defined, but its induced simplex prior changes, as in Equation 3. The same six synthetic runs and replacement gates determine whether it advances to real validation. Dropout and BN are retained as the reference recipe’s training regularization; all reported predictions use the final frozen evaluation-mode model.

PoE completion is evaluated with its own decoder,  $p_d = \text{softmax}(\text{BN}(W\theta_d))$ , not  $\theta_d \beta$ . Exported topics are conditional prototypes  $\beta_k^* = p(w | \theta = e_k)$  including frozen decoder BN. They are normalized probabilities but not additive emission distributions. This distinction also limits how directly their motif scores can be interpreted alongside LDA.

**Within-model reduction study.** A third, separately recorded phase tests the current model’s individual choices, rather than inferring their necessity from the preceding family comparisons. Eight atomic reductions



remove context entirely ( $c = 0$ ), replace leave-one-out context by the full-document mean, fix  $c = 1$ , replace top-2 by all-topic softmax routing, linearize the evidence offset, restore the ordinary softmax latent link, remove the encoder’s second hidden layer, or remove the residual MLP altogether. The last form still has a learned nonlinear topic-attention encoder: its Gaussian mean is the evidence offset and its diagonal variance is global and learned. It is not the standard ETM MLP. The linear offset is the first-order Taylor approximation at uniform evidence,  $o_d = (K/2)(r_d - 1/K)$ , fixing its scale without tuning.

Each candidate is initially fitted at  $K = 128$  with seeds 11, 23 and 37; candidates meeting the original reference criteria repeat at  $K = 36$ . The paired ETM recipe remains fixed. Exactly 90% motif retention meets the original inclusive criterion; numerical equality is checked within  $10^{-12}$  to avoid the floating-point artifact  $0.9(50/3) > 15$ . Every candidate passing both synthetic settings advances to real validation, not merely the most favorable candidate. Predeclared combinations test document context with a shallow encoder and with fixed  $c$ ; if the latter passes, the three-way combination is also tested. No-context top-1 combinations are contingent on the no-context screen passing. All negative fits remain in the inventory. The initial plan required two more training initializations after a joint real pass. The subsequent clarification below replaces automatic threshold-driven advancement with an explicitly recorded scientific shortlist, while still requiring same-seed repeats for a recommended replacement. This is a bounded deletion study, not an exhaustive architecture search or external replication.

After observing weak recovery with all-topic unit-temperature cosine routing, one additional smooth control uses logits  $\sqrt{L} \cos(h_{dw}, \hat{\alpha}_k)$ , where  $L$  is the fixed embedding dimension. Independent isotropic unit vectors have cosine variance  $1/L$ , motivating this fixed contrast normalization. This avoids attributing the unit-temperature failure solely to the absence of hard truncation. It is a declared follow-on, not an unreported temperature sweep or a claim to reproduce a published attention architecture. It receives all six synthetic fits and the unchanged advancement gates.

After fixed  $c = 1$  retained synthetic recovery, a second declared follow-on removes two-way local weighting as well: fixed context weight and hard top-1 assignment. Hard routing has zero gradient almost everywhere; the residual MLP and embedding decoder remain trainable, and no straight-through estimator is introduced. This candidate receives all six synthetic fits before real validation.

**Decision clarification during development.** The first one-layer real fit preserved chemical breadth but increased NLL by 2.62% versus the frozen reference, just outside the original 2% tolerance. Before its repeat results, the research brief explicitly accepted that scale of predictive trade-off and clarified that all arbitrary thresholds should be reference points, not rigid decision laws. The original flags remain reported; they neither prove necessity nor automatically select a winner. Descriptive one-layer repeats use seeds 7012 and 7024 with matched current controls. Specific further repeat choices are recorded before their results.

Three additional variants therefore receive complete real evaluation despite missing the synthetic reference flags: document context plus fixed  $c$  plus one layer (14.33 recovered motifs versus the numerical target of 15), no context, and ordinary softmax. The latter two also receive their previously unscheduled three  $K = 36$  fits. They test respectively a major computational deletion and restoration of ETM’s published simplex construction. This is disclosed adaptive validation development, not a preregistered confirmatory study. Recommendation weighs actual effect sizes, repeat-seed stability and the mechanisms removed; no replacement cutoff is chosen to force acceptance. A fixed capacity check also uses one 100-unit hidden layer, borrowing the width from the reviewed Pyro example without claiming to reproduce ProdLDA. It receives six synthetic fits and real validation regardless of the original flags. No width sweep is performed. Topic vectors retain identical seeded initialization across widths; differently shaped neural layers use native initialization. Document-context-only also enters the repeat shortlist after its initial fit retains chemical breadth and completion NLL. The attention-only form receives the same two repeats after its initial run reveals a major parameter reduction and a smaller, explicit chemical-inventory trade-off. The research brief subsequently accepts MLP removal while requesting a future replaceable raw-spectrum encoder. One final nested candidate combines attention-only inference, full-document context and fixed  $c = 1$ ; it receives six synthetic fits and complete seed-42 real validation regardless of the old flags. This tests whether context deletions survive MLP removal, rather than assuming that earlier MLP-based ablations transfer. No further architecture search is undertaken in this review. Before the remaining repeat results, the research brief revisits the MLP-free preference and retains an MLP-based model as the main publication candidate, with no-MLP variants as ablations. This changes the intended publication framing, not the observed data or the experiment inventory. An MLP’s presence is not itself a novelty claim; its capacity must be justified by the measured trade-offs, as discussed below.

### 3.6 Evaluation measures

For additive models, document-completion negative log-likelihood scores withheld in-vocabulary token copies after observing the other half of each spectrum:

$$\text{NLL}_{\text{comp}} = - \frac{\sum_{d,w} c_{dw}^{\text{held}} \log \left( \sum_k \hat{\theta}_{dk}^{\text{obs}} \beta_{kw} \right)}{\sum_{d,w} c_{dw}^{\text{held}}}. \quad (22)$$

Lower values indicate better prediction of unseen spectral evidence. NLL is in nats per held-out pseudo-token; a difference  $\Delta$  corresponds to a perplexity ratio  $\exp(\Delta)$ . A percentage increase in NLL is not the same percentage loss of predictive probability. For example, the one-layer seed-7043 difference is 0.233 nats versus the paired current model and 0.249 nats versus the frozen reference: 26.3% and 28.3% higher perplexity, respectively, despite NLL increases of only 2.45% and 2.62%. The reduction review also reports eligible-motif counts at inclusive SOS cutoffs 0.5, 0.6, 0.7, 0.8 and 0.9, using the same annotations and associations. This descriptive sensitivity analysis does not change the established useful-motif definition or select a favorable replacement cutoff.

Per-spectrum sparsity is measured by exact support size and the entropy-effective topic count

$$K_{\text{eff}}(\theta_d) = \exp \left[ - \sum_k \theta_{dk} \log \theta_{dk} \right]. \quad (23)$$

Writing  $\bar{\theta}_k = D^{-1} \sum_d \theta_{dk}$ , the corpus-effective inventory is  $\exp[-\sum_k \bar{\theta}_k \log \bar{\theta}_k]$ . Global inventory use is also summarized by the number of topics that are the largest mixture component for at least one spectrum (unique top-1 topics), topics with mean mixture weight above 0.0005, maximum mean topic weight and cosine similarity between topic-word probability vectors. A duplicate component is catastrophic only if at least half of the 1,000 topics join one connected component in the cosine-similarity graph at threshold 0.999.

Chemical utility is evaluated with the MS2LDA 2.0 Mass2Motif Annotation Guidance (MAG) and substructure overlap score (SOS) framework [2]. Every Spec2Vec-library row sharing a validation or test connectivity InChIKey is removed before constructing an exact inner-product vector-similarity index implemented with FAISS. The evaluated molecule therefore cannot be retrieved as its own annotation evidence.

For each topic, the 20 highest-probability fragment or neutral-loss words form a Mass2Motif pseudo-spectrum: a spectrum-like list synthesized from the learned topic probabilities. MAG searches up to 1,000 neighbors, retains at most ten unique molecules, clusters and refines them at spectral cosine-similarity threshold 0.9, and optimizes the motif against the selected cluster. A topic is *optimized* when at least one fragment or loss survives this procedure.

Clustered MAG molecules are converted to binary MACCS structural-key fingerprints, whose bits indicate predefined molecular features [25]. A bit is retained in consensus fingerprint  $A_k$  when it appears in at least 80% of the cluster. MS2LDA 2.0 associated an evaluation molecule with an LDA topic when its posterior probability exceeded 0.5 [2]. For comparison across inference families, this study instead assigns every held-out spectrum  $i$  to its dominant full-spectrum topic,

$$k_i^* = \arg \max_k \theta_{ik}. \quad (24)$$

Absolute posterior weights need not have the same calibration across variational neural models and LDA, whereas Equation 24 gives every model exactly one association per spectrum. Exact ties are resolved by the lowest topic index. This evaluation rule does not alter the fitted topic mixture. Within each topic, repeated spectra with the same connectivity InChIKey are collapsed. Let  $C_k$  be the resulting set of unique associated molecules and  $B_i$  the fingerprint of molecule  $i$ ; then

$$s_{ki} = \frac{|A_k \cap B_i|}{|A_k|}, \quad \text{SOS}_k = \frac{1}{|C_k|} \sum_{i \in C_k} s_{ki}. \quad (25)$$

The implementation assigns  $s_{ki} = 0$  if  $A_k$  contains no retained bits. An optimized motif is *SOS-evaluable* when it has at least one associated unique molecule with a valid fingerprint. A *useful* motif is evaluable with  $\text{SOS} \geq 0.6$ , combining the intermediate and high bands defined by MS2LDA 2.0. SOS measures containment of a consensus fingerprint; it is not an atom mapping, fragmentation-pathway proof or expert-confirmed annotation.

## 4 Results

### 4.1 Context improves sparse recovery in the original ETM ablation

The  $K = 36$  experiment is a controlled comparison of the two posterior changes. Starting from the channel-balanced ETM with softmax, adding contextual top-2 evidence while retaining softmax improves both planted

topic-word ( $\beta$ ) and document-mixture ( $\theta$ ) recovery, improves completion NLL, and increases the number of planted topics used as document winners. Replacing softmax with 1.5-entmax without contextual evidence does create sparse mixtures, but it reduces both recovery measures and uses fewer planted components. Thus sparsity alone is not the successful mechanism.

Combining contextual evidence with 1.5-entmax yields the strongest mean  $\beta$  and  $\theta$  recovery, retains a median of 2.00 effective topics per spectrum, and uses 13.00 distinct topics as winners, consistent with the one-to-three motif simulator. Its completion NLL is slightly worse than the context-plus-softmax variant, so the evidence supports complementarity rather than uniform dominance: context preserves and disambiguates the topic inventory, while entmax converts that evidence into exact sparse support.

Table 2:  $K = 36$  synthetic results averaged over seeds 11, 23 and 37. NLL is held-out completion negative log-likelihood (lower is better). Beta and theta are cosine recovery of the planted topic-word and document-topic matrices; median eff. is the median entropy-effective topic count. Active topics have mean corpus weight above 0.005, and unique top-1 counts distinct document winners.

| Formulation                         | NLL    | Beta   | Theta  | Median eff. | Active | Unique top-1 |
|-------------------------------------|--------|--------|--------|-------------|--------|--------------|
| Balanced ETM + softmax              | 6.4710 | 0.3052 | 0.4869 | 3.59        | 8.00   | 6.67         |
| Balanced ETM + 1.5-entmax           | 6.5982 | 0.2585 | 0.4005 | 1.92        | 5.33   | 5.00         |
| Contextual top-2 evidence + softmax | 6.2460 | 0.4427 | 0.6990 | 3.27        | 13.00  | 11.33        |
| <b>Contextual Sparse ETM</b>        | 6.2870 | 0.4646 | 0.7535 | 2.00        | 14.33  | 13.00        |

The high-overcompleteness experiment strengthens this interpretation. At  $K = 128$ , entmax without context has median exact support 2, recovers 2 of the 18 planted motifs at matched beta cosine at least 0.50, and uses only 3 learned topics as document winners. Contextual Sparse ETM retains median exact support 2 while recovering 18 planted motifs and using 19 learned topics as winners. It also gives the strongest beta/theta recovery (0.6643/0.9465), dominant-motif accuracy 0.9313 and the lowest completion NLL among the three high- $K$  formulations.

Table 3: Seed-11 synthetic stress with 18 planted motifs and 128 fitted topics. NLL is completion negative log-likelihood; beta and theta are planted-matrix recovery; accuracy compares the dominant fitted and planted motif. Eff. is the entropy-effective topic count, recovered is the number of planted motifs whose truth-matched beta cosine is at least 0.50, support counts exactly non-zero document-topic weights, and unique counts distinct learned document winners.

| Formulation                  | NLL    | Beta   | Theta  | Accuracy | Recovered | Eff. | Support | Unique |
|------------------------------|--------|--------|--------|----------|-----------|------|---------|--------|
| Balanced ETM + 1.5-entmax    | 6.7375 | 0.2344 | 0.2965 | 0.2625   | 2         | 1.72 | 2.0     | 3      |
| Balanced ETM + softmax       | 6.4175 | 0.3696 | 0.5782 | 0.5437   | 6         | 3.67 | 128.0   | 7      |
| <b>Contextual Sparse ETM</b> | 6.1254 | 0.6643 | 0.9465 | 0.9313   | 18        | 1.40 | 2.0     | 19     |

## 4.2 Contextual Sparse ETM expands the chemically assessable motif inventory

Figure 3 and Table 4 summarize the held-out MS<sup>n</sup>Lib comparison. Before repeated connectivity keys are collapsed within topics, every method contributes exactly 7777 dominant-topic spectrum associations. The comparison therefore gives each model the same number of opportunities for molecule association, independent of absolute posterior calibration. The three ETM rows trace the model lineage from canonical ETM, through channel balancing, to the complete contextual sparse posterior.

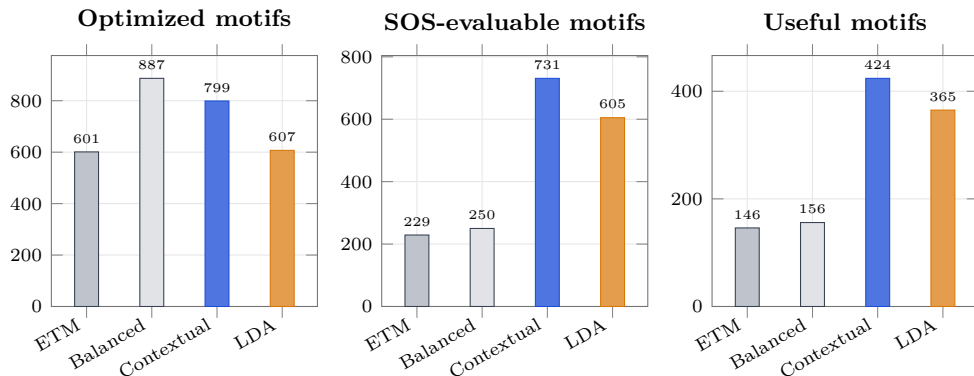


Figure 3: Held-out test motif inventory after successive chemical filters. ETM is the canonical Embedded Topic Model with fixed train-only spectral-word coordinates and dense softmax document mixtures. Balanced is the same ETM with separate 50/50 fragment and neutral-loss decoder normalization. Contextual is the complete Contextual Sparse ETM, which adds contextual top-2 posterior evidence and 1.5-entmax to Balanced. LDA is Tomotopy 0.13.0 latent Dirichlet allocation. Optimized means that Mass2Motif Annotation Guidance (MAG) retains at least one fragment or loss; SOS-evaluable means that the substructure overlap score can be computed from at least one associated molecule; useful means SOS at least 0.6. All four models use  $K = 1000$ , the same training split, the same held-out test split and the same chemical-evaluation protocol. Each test spectrum is associated with its single dominant full-spectrum topic, giving every model exactly 7777 spectrum–topic associations before connectivity-key deduplication. The three ETM bars use training seed 7043; Table 5 reports the other Contextual Sparse ETM seeds.

Table 4: Final held-out test comparison under one dominant-topic association per spectrum. SOS is the substructure overlap score, useful/evaluable is the fraction of evaluable motifs with SOS at least 0.6, and NLL is held-out completion negative log-likelihood, for which lower is better. Bold values are the numerically strongest entry in each metric; no single metric defines overall model quality. The ETM rows use training seed 7043; same-split seed sensitivity for Contextual Sparse ETM is reported in Table 5.

| Model                        | Optimized  | Evaluable  | Useful     | Useful/evaluable | Mean SOS      | NLL ↓         |
|------------------------------|------------|------------|------------|------------------|---------------|---------------|
| Canonical ETM                | 601        | 229        | 146        | <b>63.8%</b>     | 0.6307        | <b>8.6860</b> |
| Fragment/loss-balanced ETM   | <b>887</b> | 250        | 156        | 62.4%            | 0.6369        | 8.7797        |
| <b>Contextual Sparse ETM</b> | 799        | <b>731</b> | <b>424</b> | 58.0%            | 0.6244        | 9.5355        |
| Tomotopy LDA                 | 607        | 605        | 365        | 60.3%            | <b>0.6379</b> | 9.7348        |

Channel balancing alone raises the optimized count from 601 to 887, but evaluable motifs increase only from 229 to 250 and useful motifs from 146 to 156. The complete contextual sparse posterior instead yields 731 evaluable and 424 useful motifs. This contrast shows that a larger optimized inventory alone does not account for the final gain.

Relative to Tomotopy, Contextual Sparse ETM produces 20.8% more evaluable motifs and 16.2% more useful motifs, while reducing completion NLL from 9.7348 to 9.5355. The gain is not uniform across metrics: Tomotopy’s conditional mean SOS is higher by 0.0135 (0.6379 versus 0.6244), and its useful/evaluable fraction is higher by 2.3 percentage points (60.3% versus 58.0%). Canonical and balanced ETM retain still lower completion NLL and higher conditional mean SOS, but they produce substantially fewer evaluable and useful motifs.

Figure 4 separates the two chemical axes. Contextual Sparse ETM expands the evaluable inventory, whereas Tomotopy retains the higher average SOS within its smaller, model-specific evaluable set.

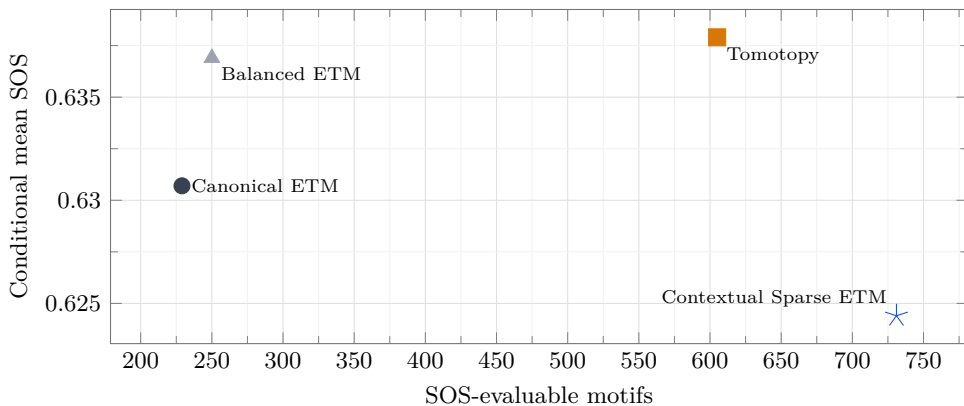


Figure 4: Discovery breadth versus conditional chemical quality on the held-out test split. Horizontal position counts motifs for which the substructure overlap score (SOS) can be evaluated; vertical position averages SOS only over that model-specific set. The axes therefore answer complementary questions rather than forming a single combined score. Points compare canonical and fragment/loss-balanced Embedded Topic Models (ETMs), the complete Contextual Sparse ETM and Tomotopy LDA. The ETM points use training seed 7043; the seed sensitivity of Contextual Sparse ETM is reported separately in Table 5.

Changing only initialization and minibatch order tests whether this discovery gain depends on one fitted checkpoint. The three runs give 798–805 optimized, 731–740 evaluable and 415–432 useful motifs. Mean SOS spans 0.6182–0.6244 and completion NLL spans only 9.5161–9.5355. Median effective topics remain within 3.68–3.74. Across the runs, 914–922 topics become the largest component for at least one test spectrum. No run contains a catastrophic duplicate component.

Table 5: Same-split training-seed stability. SOS is the substructure overlap score, NLL is completion negative log-likelihood and median eff. is the median entropy-effective topic count; unique counts distinct document winners. The final row reports the descriptive sample mean and standard deviation for three initializations, not population uncertainty.

| Seed          | Optimized       | Evaluable       | Useful          | Mean SOS            | Median SOS          | NLL                 | Median eff.       | Unique          |
|---------------|-----------------|-----------------|-----------------|---------------------|---------------------|---------------------|-------------------|-----------------|
| 7043          | 799             | 731             | 424             | 0.6244              | 0.6222              | 9.5355              | 3.680             | 917             |
| 23            | 798             | 740             | 432             | 0.6213              | 0.6228              | 9.5261              | 3.714             | 922             |
| 37            | 805             | 740             | 415             | 0.6182              | 0.6198              | 9.5161              | 3.744             | 914             |
| Mean $\pm$ SD | 800.7 $\pm$ 3.8 | 737.0 $\pm$ 5.2 | 423.7 $\pm$ 8.5 | 0.6213 $\pm$ 0.0031 | 0.6216 $\pm$ 0.0016 | 9.5259 $\pm$ 0.0097 | 3.713 $\pm$ 0.032 | 917.7 $\pm$ 4.0 |

The NLL, sparsity and chemical-score ranges test whether the mechanism is specific to one initialization. Their joint variation motivates reporting motif counts together with the SOS distribution and predictive fit rather than reducing performance to one average.

### 4.3 Sparse spectrum assignments coexist with broad corpus-level topic use

The representative run has median 3.68 entropy-effective topics and median exact support 6 per test spectrum; 95% of spectra have support at most 13. At corpus level, 917 of 1,000 topics win at least one spectrum, 537 exceed mean usage 0.0005 and the corpus-effective topic count is 549.3. The largest connected component at topic–word cosine similarity  $\geq 0.999$  contains only 1 topics.

Table 6: Held-out test sparsity and inventory diagnostics. Eff. is the entropy-effective topic count; support counts exactly non-zero mixture weights; unique top-1 counts distinct document winners; and active topics have mean weight above 0.0005. Corpus eff. applies the entropy-effective count to mean corpus weights, and nearest beta is mean nearest-neighbor cosine similarity between topic–word probability vectors.

| Model                        | Median eff. | Median support | Unique top-1 | Active | Corpus eff. | Nearest beta |
|------------------------------|-------------|----------------|--------------|--------|-------------|--------------|
| Canonical ETM                | 41.78       | 1000.0         | 318          | 300    | 345.9       | 0.308        |
| Fragment/loss-balanced ETM   | 43.83       | 1000.0         | 279          | 280    | 365.7       | 0.396        |
| <b>Contextual Sparse ETM</b> | 3.68        | 6.0            | 917          | 537    | 549.3       | 0.150        |

The ETM baselines retain dense spectrum-level mixtures and use fewer distinct document winners. Contextual Sparse ETM reverses both properties: individual spectra are sparse, yet 917 topics receive a top-1 assignment



somewhere in the test set. Its maximum mean topic usage is only 0.0183, and mean nearest-topic topic–word cosine similarity is 0.1499, providing additional evidence that sparsity does not come from a single dominant or globally duplicated topic group.

Computationally, the 120-epoch optimization loop takes  $13.6 \pm 0.1$  minutes. With the trained model already resident on the GPU, deterministic inference processes 22,672 full test spectra per second. Peak memory actively allocated by PyTorch and memory reserved by its CUDA cache are 0.907 and 1.116 GB, respectively. Peak process resident memory is 2.876 GB, while at least 11.81 GB of system memory remains available.

These measurements show that the method fits comfortably on the evaluated GPU system. They are implementation and hardware measurements rather than algorithm-independent complexity claims, and runtime is not used for the scientific model ranking.

#### 4.4 Simplification requires breadth as well as sparse mixtures

The independent review does not identify a replacement among the initial ETM simplifications. Although all three candidates pass the paired synthetic screen (Table 10), synthetic success does not guarantee real chemical breadth. Fixed-scale BN–entmax reaches 4.23 effective topics per validation spectrum, but only 80 topics win any spectrum, 67 become evaluable and 42 are useful (Table 7). Sparse assignments here coexist with severe underuse of the topic inventory. The strict near-duplicate flag alone does not diagnose that failure.

Removing channel balancing retains broad topic use (827 winners) and compact mixtures (3.80 effective topics), but loses chemical breadth: 524 evaluable and 322 useful motifs versus the paired current model’s 675 and 394. Thus balancing is not justified solely by an appealing domain story; this ablation provides direct evidence for retaining it in the complete model. The historical and paired current references agree on the decision despite their batch-size difference. Neither candidate qualifies for confirmatory seed repeats under the recorded joint criteria.

| Model                          | Evaluable | Useful | Mean SOS | NLL   | Eff. topics | Winners |
|--------------------------------|-----------|--------|----------|-------|-------------|---------|
| ETM (historical)               | 199       | 118    | 0.635    | 8.677 | 40.46       | 272     |
| Balanced ETM (historical)      | 228       | 153    | 0.647    | 8.775 | 45.09       | 254     |
| Contextual (historical)        | 672       | 400    | 0.636    | 9.515 | 3.66        | 838     |
| Tomotopy (historical)          | 573       | 359    | 0.642    | 9.689 | —           | —       |
| Contextual (paired)            | 675       | 394    | 0.623    | 9.531 | 3.72        | 829     |
| BN–entmax, fixed scale         | 67        | 42     | 0.627    | 9.146 | 4.23        | 80      |
| Contextual, no channel balance | 524       | 322    | 0.629    | 9.582 | 3.80        | 827     |

Table 7: Validation-only initial simplification review,  $K = 1000$ . Useful means SOS at least 0.6; mean SOS is conditional on evaluability. NLL is completion negative log-likelihood per in-vocabulary token; effective topics are the median exponentiated mixture entropy; winners count topics with at least one dominant assignment. Historical neural references use batch 256; the three new rows use batch 200 with matched seed 7043. Dashes are unreported historical diagnostics, not zeros. No new test evaluation occurs.

The preceding BN/prior study gives the same warning from another direction (Table 11). Its best useful-count variant, fixed-scale BN, produces 296 useful motifs and good completion NLL (8.231), but a median of 635 effective topics. Adding the Laplace-approximated Dirichlet prior does not simultaneously restore compactness and breadth. Learned-scale BN plus that prior instead becomes compact while serving only 56 winning topics. All four displayed results use final training-only normalization calibration; unreliable lagging-statistic pilots are retained in the evidence, not selected as favorable comparisons.

**Going beyond ETM ablations.** The expanded screen tests four published-family recipes and one explicitly staged enhancement (Table 12). At  $K = 128$ , stock ProdLDA recovers all 18 planted motifs across all three seeds, but its mean median effective-topic count is 59.46. Replacing only its latent softmax with entmax retains all 18 motifs and reduces that count to 4.34; mean beta and theta recovery are 0.834 and 0.923. This is strong synthetic evidence for an understandable alternative, not grounds to dismiss it from literature alone. The direct-Dirichlet recipes instead retain roughly 104–111 effective topics at high  $K$ ; the additive variants recover no planted motif at cosine 0.50 within this fixed training budget. Their sparse prior does not deliver a compact inferred mean.

The real validation results resolve the replacement question differently (Table 8). Stock ProdLDA has 846 winning topics and 332 useful motifs, but a median of 657.38 effective topics. Its entmax extension is compact (2.84 effective topics) and has better completion NLL than the current model (9.122 versus 9.531), yet serves

only 207 winning topics and yields 190 evaluable/118 useful motifs. Neither passes the joint gate. High MAG optimization coverage is insufficient: 900 entmax-ProdLDA prototypes are optimized, but only 190 qualify for the molecule-associated SOS score. The loss of breadth therefore cannot be hidden by reporting optimization coverage alone.

| Model                         | Evaluable | Useful | Mean SOS | NLL   | Eff. topics | Winners |
|-------------------------------|-----------|--------|----------|-------|-------------|---------|
| Contextual (paired)           | 675       | 394    | 0.623    | 9.531 | 3.72        | 829     |
| ProdLDA (Pyro-reference port) | 537       | 332    | 0.636    | 9.004 | 657.38      | 846     |
| ProdLDA + entmax              | 190       | 118    | 0.629    | 9.122 | 2.84        | 207     |

Table 8: Published-family real validation,  $K = 1000$ , seed 7043. Column definitions and the common spectrum-association rule follow Table 7. ProdLDA rows use their actual PoE decoder for NLL and one-hot conditional topic prototypes for chemical evaluation; they are not scored as additive LDA emissions. Recipes differ in neural width and optimization settings as stated in Section 3.5.

The first two phases of the independent review add 54 synthetic fits and five real-validation fits, while preserving the earlier 53 experiment records. None of the new models is evaluated on test data. No simpler tested recipe meets the full replacement criterion, and none is promoted or integrated into production. This supports a bounded retention decision, not an assertion that the current form is the globally simplest possible neural model.

#### 4.5 A simpler contextual form preserves the ETM backbone

The completed third phase changes that initial retention decision. Replacing leave-one-out context by the whole-spectrum mean is the best-supported simplification under the publication brief: it retains the conventional ETM MLP, Gaussian recognition model, embedding decoder and latent-space ELBO, while removing token-specific context subtraction and division. Table 9 reports three paired training seeds on the same validation split.

| Model                 | Parameters | Evaluable        | Useful           | Mean SOS          | NLL               | Eff. topics     |
|-----------------------|------------|------------------|------------------|-------------------|-------------------|-----------------|
| Current               | 19,278,001 | $666.3 \pm 7.5$  | $389.7 \pm 17.9$ | $0.624 \pm 0.004$ | $9.540 \pm 0.011$ | $3.76 \pm 0.03$ |
| Full-document context | 19,278,001 | $665.7 \pm 7.6$  | $390.7 \pm 2.1$  | $0.629 \pm 0.002$ | $9.525 \pm 0.023$ | $3.72 \pm 0.03$ |
| Evidence-only encoder | 49,001     | $546.0 \pm 21.9$ | $341.3 \pm 25.7$ | $0.641 \pm 0.009$ | $9.115 \pm 0.010$ | $4.38 \pm 0.02$ |
| One hidden layer      | 18,637,201 | $668.7 \pm 10.4$ | $400.7 \pm 6.1$  | $0.628 \pm 0.006$ | $9.772 \pm 0.009$ | $3.61 \pm 0.01$ |

Table 9: Within-model validation repeats at  $K = 1000$ : means  $\pm$  sample standard deviations over seeds 11, 23 and 42 (training seeds offset by 7001). Current means the original two-layer, width-800 MLP with leave-one-out context; one hidden layer has width 800. Parameters count trainable coordinates, not fixed SGNS embeddings. Useful means SOS at least 0.6, conditional mean SOS describes evaluable motifs, NLL is completion loss in nats per token, and effective topics are the median exponentiated mixture entropy per run. The seed-42 current run is reused from phase 1, not retrained or counted twice. Three initializations describe same-split variability, not external confidence bounds.

Full-document context yields 390.7 useful motifs on average versus 389.7 for the current model, with 665.7 versus 666.3 evaluable motifs and 3.72 versus 3.76 effective topics. NLL is lower in all three paired runs, averaging 9.525 versus 9.540. Useful-count differences are  $-15$ ,  $+23$  and  $-5$ : the evidence supports preserved usefulness, not a consistent chemical gain. Its smaller observed useful-count SD is descriptive with only three seeds. One run has 799 winning topics, one below the old 800 reference flag; that does not overturn the substantive comparison. At SOS cutoffs 0.7 and 0.8, full-document context yields more qualifying motifs in every paired seed (Table 15). The conclusion is not created solely by the 0.6 cutoff. These are correlated sensitivity summaries, not independent replications. Parameter count is unchanged, and concurrent training queues do not provide a fair speed benchmark.

**MLP capacity is a trade-off, not a novelty requirement.** A single 800-unit layer retains chemical breadth: 400.7 useful motifs versus 389.7, with 3.61 effective topics. Mean NLL rises by 0.232 nats per token, equivalent to about 26.1% higher perplexity; describing this only as a 2.43% NLL increase would understate its predictive meaning. This is a valid shallower alternative if that trade-off is acceptable, not an automatic failure

at the former 2% cutoff. It removes only 3.3% of trainable parameters because the vocabulary-to-hidden and topic-output layers dominate.

The single 100-unit-layer run removes 87.7% of trainable parameters and produces 377 useful motifs versus the paired current model’s 394, with NLL 9.680 versus 9.531 and 3.65 effective topics. That is 4.3% fewer useful motifs and approximately 16.1% higher perplexity, not a failed model merely because it has 790 winners. Its strong synthetic recovery and near-preserved chemistry make it a credible future capacity candidate. Only one real seed was scheduled, so it is not promoted over the repeated full-document result. No combined full-document/100-unit model was tested. Even the tested full-document plus 800-unit-layer combination loses usefulness (353 versus 394); successful individual deletions do not imply a successful combination.

The attention-only encoder removes 99.75% of trainable parameters and improves NLL to 9.115, but averages 341.3 useful motifs, 12.4% fewer than the current model, and 638 versus 827 winning topics. Its mean nearest-topic beta cosine is 0.435 versus 0.157, indicating more overlapping emissions without triggering the strict catastrophic-duplicate definition. The trade-off is real, not a reason to relabel it LDA or NMF. Under the latest published-lineage preference, it remains a boundary ablation rather than the publication backbone.

**Which remaining choices are justified?** The ordinary-softmax control produces 395 useful motifs and better NLL (9.125), but 47.58 effective topics versus 3.72 for the paired current model. Thus entmax is directly supported for compact spectrum explanations, not because softmax fails chemical usefulness. Removing context entirely retains 366 useful motifs (7.1% fewer) with NLL 9.575 and worse synthetic recovery. This supports context as helpful; it does not prove context is indispensable for every acceptable real-data trade-off. Fixing  $c = 1$  gives 374 useful motifs, or 379 when combined with document context; fixed- $c$  hard top-1 also gives 379, with a more compact 2.81 effective topics. These one-seed results remain defensible simpler options, but do not establish that their small chemical losses are stable. Retaining the learned scalar and top-2 routing is a conservative choice, not proof that both are absolutely necessary. Untruncated routing performs substantially worse on synthetic recovery even after its declared fixed contrast scaling (Table 13). Channel balancing has separate real ablation support from phase 1. Taken together, the evidence favors a small set of named ETM adaptations rather than an assertion that every operation is irreducible.

**Complete outcomes, including failure.** This phase contains 87 completed synthetic fits (29 three-seed groups), 20 completed real fits with full chemical evaluation, and one numerically failed real configuration. The linear evidence offset passes the synthetic screen but raises a non-finite training loss after its epoch-80 recovery checkpoint, before completing 120 epochs. No final metrics exist; it was not retuned or retried. Its exact failing epoch and numerical root cause were not instrumented. An earlier interrupted document-context execution is separately recorded and restarted from scratch; it is not a second independent seed or this numerical failure. All completed real evaluations have zero MAG exceptions and use the same 3,889 spectrum associations. Every scheduled outcome is retained in Table 14 and the evidence inventory. Across all three new phases there are 166 successful fits and one failed configuration, in addition to the 53 audited earlier experiment records. No new candidate loads test matrices. All overnight queues are complete; no additional fits are required to report these results.

## 5 Discussion

### 5.1 Mechanism and model economy

The proposed model separates two roles. The embedding decoder gives every topic an explicit fragment/loss distribution, preserving the object required for Mass2Motif inspection and MAG. The posterior adaptation asks a different question: which topics are supported by the words observed together in this particular short spectrum? Context lets a common fragment or loss contribute differently when its surrounding evidence changes; the third phase shows that the simpler full-spectrum mean is sufficient in the tested setting. Restricting each word to its two strongest topic matches prevents weak similarity to hundreds of topics from accumulating into a diffuse document signal.

The centered log offset does not replace the variational encoder. It supplies a bounded evidence correction in the encoder’s own topic coordinates. Uniform contextual evidence leaves the posterior unchanged, and topics outside local top-2 assignments retain finite evidence through the fixed  $1/K$  pseudocount. Finally, 1.5-entmax converts the corrected latent representation into exact support. The synthetic results show why these operations are complementary: contextual evidence improves recovery and topic use, while sparse projection controls the document support.

The complete model can be explained through four objects: fixed spectral-word coordinates, learned ETM topic coordinates, one contextual evidence distribution per spectrum and explicit  $\theta$  and  $\beta$  probability vectors. Channel balancing has a direct domain interpretation and no learned parameters. The posterior adaptation adds

only the context scalar  $c$ ; assignment temperature, evidence strength, top-2 support and pseudocount are fixed. Consequently, the method remains recognizable as an ETM rather than replacing the generator with a separate clustering system. One added scalar describes parameter economy, not the number of modeling assumptions or computational operations: channel normalization, leave-one-out context, top-2 matching, smoothed evidence and the sparse latent link remain distinct choices. Minimality must be assessed against simpler fitted alternatives, not inferred from parameter count alone.

The first two phases support retaining the model family; the completed third phase supports simplifying its context to the full-spectrum mean. Describe the resulting form as three explicit adaptations of published ETM—channel normalization, contextual variational evidence and the entmax latent link. Do not describe the full generator as unchanged, or equate a single extra parameter with a single modeling assumption. Published ProdLDA plus entmax is an understandable alternative, but its synthetic success does not establish sufficient real motif breadth. Direct-Dirichlet inference is also principled; its tested recipe’s failure is not evidence against that family in general. The original context ablation removes the whole evidence branch. It does not isolate leave-one-out mixing from top-2 routing and the smoothed log offset; the third phase tests these choices separately and does not establish universal necessity of every retained setting.

**What is and is not a publication contribution.** Keeping a conventional MLP cannot by itself establish methodological novelty: neural recognition networks are already part of AVITM and ETM [6, 7]. Conversely, removing the residual MLP does not turn this model into classical LDA or ordinary nonnegative matrix factorization (NMF). The reduced form still has a nonlinear learned attention encoder, embedding-constrained fragment/loss emissions, Gaussian latent uncertainty, an entmax-induced sparse mixture distribution and a variational objective. Classical LDA instead uses a Dirichlet document-mixture prior [4]. Sharing the algebraic mean reconstruction  $\theta\beta$  with other topic/factor models does not make their priors, constraints or inference procedures identical.

The defensible claim is therefore an ETM-derived spectral model whose explicit posterior geometry and sparse mixtures address the joint problem of compact spectrum explanations and broad chemically interpretable motif discovery. Neither attention nor sparse topic maps are individually new [14, 13]; the work should not claim to be the first such model without a dedicated, broader priority review. An MLP can be retained to recover chemical breadth or improve inference flexibility, and a shallower or narrower MLP may be sufficient. Those are measurable reasons to retain capacity, not reasons to add complexity for appearance. The initial matched full/no-MLP comparison has 394 versus 326 useful validation motifs, a substantive breadth difference that must be reported alongside the reduced form’s much smaller parameter count and better completion NLL. The completed repeats retain that average breadth trade-off. The recommended publication candidate uses an MLP with full-document context; no-MLP results remain informative ablations rather than being discarded for lacking a standard neural layer.

The final selection priority is proximity to a published ETM backbone, not the lowest parameter count across recognition families. Retaining the conventional MLP Gaussian encoder inherits a well-established inference design; replacing it with attention alone creates an additional design choice to defend. Width and depth reductions within an ordinary MLP can still be evaluated as capacity settings without proposing a different generative model. Published ETM provides the starting framework, not blanket justification for every adaptation: channel normalization changes the emission distribution, entmax changes the induced mixture prior, and the contextual offset changes recognition. Each departure is explicitly specified and ablated here. The publication should say “ETM with these named adaptations,” not imply that its modified generative distribution is the original published ETM unchanged. The attention- only form remains a useful boundary ablation and future option, not the recommended base under this published-lineage constraint.

## 5.2 Exact reductions and a cleaner model form

**Recommended development form.** For normalized spectral counts  $x_d$  and fixed unit word vectors  $\hat{\rho}_w$ , compute one mean  $s_d = \sum_w x_{dw} \hat{\rho}_w$  and

$$h_{dw} = \text{unit}(\hat{\rho}_w + cs_d), \quad r_d = \sum_w x_{dw} \text{softmax}_{\text{top-2}}(h_{dw} \hat{\alpha}^\top). \quad (26)$$

Use the same two-layer MLP for Gaussian parameters as before, with posterior mean  $m_d = \mu_{\text{MLP}}(x_d) + C \log(1 + Kr_d)$ , diagonal MLP variance, entmax mixtures, channel-balanced ETM emissions and the Gaussian latent-space ELBO. The only empirical change in the recommended variant is full-document rather than leave-one-out context; the log1p rewrite below is algebraic. This remains neural and ETM-derived, with a single learned context scalar. The implementation is isolated as `reduced_document_context`; neither the frozen historical model nor the production Tomotopy path is overwritten.

Some apparent complications can be removed algebraically, without a favorable ablation score. Let  $C = I - \mathbf{1}\mathbf{1}^\top/K$  centre a topic vector. The posterior evidence is exactly

$$o_d = C \log(r_d + \mathbf{1}/K) = C \log(\mathbf{1} + Kr_d). \quad (27)$$

This exposes a single bounded evidence correction. It does not eliminate the fixed choice of unit total smoothing mass. Setting the pseudocount to zero while retaining sparse routing and a log offset would produce infinite Gaussian means whenever  $r_{dk} = 0$ , and hence no finite Gaussian KL. A different finite offset, such as the tested linear form, is a meaningful alternative; an unqualified log 0 ablation is not.

For  $c = 0$ , attention depends only on the vocabulary word, so the complete evidence branch becomes  $r_d = x_d A$ , where  $A_{wk}$  is the top-2 softmax of word–topic cosine similarities. No document-context calculation or trainable context scalar remains. If softmax replaces entmax, the deterministic posterior has the particularly simple form

$$\hat{\theta}_{dk} \propto \exp(\mu_{dk})(1 + Kr_{dk}). \quad (28)$$

This is a multiplicative evidence correction to conventional ETM inference; whether it performs adequately is an empirical question, not an algebraic one.

Even with leave-one-out context retained, its direction has a compact form. Writing  $s_d = \sum_v x_{dv} \hat{\rho}_v$ , for  $x_{dw} < 1$ ,

$$h_{dw} = \text{unit}([1 - (1 + c)x_{dw}]\hat{\rho}_w + cs_d). \quad (29)$$

The positive factor  $1 - x_{dw}$  cancels in the final normalization. Singleton words use  $h_{dw} = \hat{\rho}_w$ , as in the original definition. This is an exact directional identity away from numerical norm-clamping boundaries, not the removal of context. It also explains the full-document approximation: that ablation replaces the word-dependent coefficient  $1 - (1 + c)x_{dw}$  by one. The difference can matter for a dominant peak, even when it is small for low-weight words. The original guarded numerical implementation is preserved.

Channel balancing also admits a simpler conditional interpretation. For channel  $g \in \{\text{fragment}, \text{loss}\}$ , define  $\beta_{kv}^{(g)} = \text{softmax}_{v \in g}(\alpha_k^\top \rho_v)$ . The likelihood is an ETM conditional on the observed channel,

$$p(v \mid \theta_d, g) = \sum_k \theta_{dk} \beta_{kv}^{(g)}, \quad p(v, g \mid \theta_d) = \frac{1}{2} p(v \mid \theta_d, g). \quad (30)$$

For the token reconstruction term used here, the balanced negative log-likelihood equals its conditional counterpart plus  $N_d \log 2$ , a parameter-independent constant. Thus the half-mass factor need not be described as another learned mechanism. It must still be restored when reporting joint word probabilities and comparing NLL with single-vocabulary models. Conditional independence of fragment/loss tokens remains a modeling approximation, not a chemical identity. The shared-mixture, channel-specific-emission structure follows an established multi-view topic-model pattern, exemplified by polylingual topic models [8]. Here the two views are fragment and loss vocabularies, with their emissions tied through ETM embeddings; the sparse Gaussian-derived mixture prior and neural inference are different from that model’s Dirichlet construction. Thus a two-channel conditional ETM is a defensible way to explain the decoder without presenting the factor of one half as an independent learned enhancement.

Finally, write  $m_d = \mu_d + o_d$  and retain the same diagonal variance. Replacing the Gaussian mean by  $Cm_d$  leaves the entire induced distribution of  $\theta_d$  unchanged: for every shared Gaussian noise draw, both simplex links are invariant to subtracting  $\bar{m}_d \mathbf{1}$ . However,

$$\text{KL}(q_{m_d} \| p) - \text{KL}(q_{Cm_d} \| p) = \frac{K}{2} \bar{m}_d^2 \geq 0. \quad (31)$$

This removes a redundant posterior-mean direction and weakly tightens the latent ELBO without altering predictions. It is implementable by subtracting the across-topic mean from each column of the mean head’s weight matrix and centring its bias in a fitted checkpoint. It is not a license to delete evidence centring alone: uncentred offsets generally increase the KL even though the simplex projection ignores their common shift. Nor does this argument eliminate the variances or preserve training trajectories after reparameterization. The audit verifies these identities on all validation spectra, using double precision and the frozen float32-trained weights; no new fit or test-spectrum access is involved. The removable penalty averages only 0.011 nats per spectrum against a mean Gaussian KL of 322.12 nats, so this is chiefly a conceptual simplification, not evidence of a substantial performance gain. The maximum identity error is below  $3 \times 10^{-13}$  in this audit.

Interpretability nevertheless remains statistical. A high-probability fragment/loss pattern is a Mass2Motif hypothesis, and SOS measures containment of a MAG-derived consensus fingerprint. Chemical naming, atom-level mapping and fragmentation-mechanism claims still require expert review and, ideally, orthogonal experimental evidence.



### 5.3 A clean boundary for future direct-spectrum encoders

**The MLP-free neural form.** Using the same evidence  $r_d$ , channel-balanced  $\beta$  and Gaussian-entmax generative model, the attention-only posterior is simply

$$m_d = C \log(\mathbf{1} + K r_d), \quad q(z_d | x_d) = \mathcal{N}(m_d, \text{diag}(e^v)), \quad v \in \mathbb{R}^K. \quad (32)$$

The learned topic vectors and context weight make the mean a nonlinear amortized recognition function; a dense MLP is not required for it to be neural. The ordinary latent-space negative ELBO remains expected raw-count reconstruction plus analytic Gaussian KL with unit weight. The same learned topic vectors can appear in the generator and recognition function: this parameter sharing does not invalidate the variational bound. At  $K = 1000$  and embedding dimension 48, trainable state is  $48K + K + 1 = 49,001$  values, versus 19,278,001 in the matched complete model. The fixed  $21,233 \times 48$  SGNS table contributes another 1,019,184 coordinates and its pretraining remains part of the pipeline. This is a 99.75% reduction in trainable parameters, not an equal reduction in total memory or runtime. This deletion also changes document-dependent posterior variance to a global learned diagonal variance. It supports the complete reduced form, not an isolated claim that the mean residual or variance network individually has no value. The nested document-context/fixed-weight variant removes the one scalar as well; its separate fits test those additional changes.

Removing the residual MLP does not remove neural inference or require a permanent commitment to a bag-of-words encoder. The research implementation separates an input encoder from the Gaussian latent prior, analytic KL, entmax mixture map and embedding-based emission decoder. Its only posterior contract is

$$\text{Encoder}(\text{observations}; \alpha, \rho) \longrightarrow (m_d, \log \sigma_d^2) \in \mathbb{R}^K \times \mathbb{R}^K. \quad (33)$$

The core does not inspect or tokenize the observations. Today’s attention encoder consumes normalized spectral-word weights and uses the supplied geometry; a future encoder may consume peak arrays or spectrum embeddings and ignore the word vectors. Shared topic vectors have one parameter owner, so decoder and encoder gradients can act on the same geometry without duplicated trainable state. The default shell imports the fitted MLP-free checkpoint without changing its predictions or gradients; it is an implementation boundary, not another fitted model or a new enhancement.

DreaMS is a concrete future option, not part of the present experiments. Its published work and official interface describe self-supervised spectrum representations and a 1024-dimensional vector per spectrum [9, 10]. A later study could compare a frozen DreaMS representation plus a small Gaussian-posterior head with end-to-end fine-tuning. Such a head is an explicit adaptation, not the removed large vocabulary-to-hidden MLP. The present work adds neither a DreaMS dependency nor pretrained weights, and a toy peak-input contract test is not evidence of DreaMS performance. The official convenience embedding API currently loads a contrastively fine-tuned head, so a future experiment must distinguish that checkpoint from the self-supervised backbone and audit both training stages. Peak selection and spectrum preprocessing also remain: avoiding MS2LDA’s word binning does not mean using unprocessed measurements.

There are two scientifically different extensions. Replacing only the encoder can avoid MS2LDA word binning on the inference input, while retaining token targets for training and the existing fragment/loss motif distributions. Removing tokenization from the entire model would additionally require a continuous or peak-set emission likelihood, a treatment of variable peak counts and precursor information, and a new motif-export/chemical-evaluation contract. The latter is a new observation model, not an encoder swap, and is explicitly deferred. A foundation-model study must also audit overlap between its pretraining corpus and evaluation spectra/compounds; it cannot inherit the current train-only SGNS leakage claim. These questions require new untouched data and do not alter this validation-only review.

### 5.4 Breadth–quality trade-offs and generalization priorities

Optimized, evaluable and useful motif counts measure how much of the learned inventory becomes available for chemical interpretation. Mean and median SOS measure chemical containment conditional on passing the evaluability filter. A model can therefore expose more useful motifs in absolute terms while having a lower average over that larger set. Contextual Sparse ETM exhibits this trade-off relative to Tomotopy: it increases evaluable and useful counts, while Tomotopy has conditional mean SOS higher by 0.0135 and a useful/evaluable fraction higher by 2.3 percentage points. The breadth–quality plot makes this conditional denominator explicit and avoids treating an average over one model-specific motif set as though it summarized the same set for every model.

Completion NLL provides a third axis. The proposed model predicts withheld evidence better than Tomotopy but not as well as the dense ETM baselines. This may reflect the stronger allocation constraint imposed by sparse document mixtures, although the present experiments do not isolate sparsity as the cause of the NLL difference. The results therefore do not support a blanket superiority claim. They support the narrower conclusion that the



posterior adaptation materially increases chemically assessable motif breadth while preserving sparse assignments and a broad global inventory.

These axes correspond to different uses. Completion NLL tests probabilistic generalization, support diagnostics test whether spectrum explanations remain compact, and the MAG and SOS measures determine how much chemically interpretable material reaches a downstream analyst. For discovery, the larger useful set is valuable even when it should subsequently be ranked or filtered by conditional chemical evidence. Reporting all three prevents a favorable result on one axis from concealing collapse on another.

The next decisive experiment is generalization beyond the present frozen split. A second scaffold split or an external spectral library would test sensitivity to data composition. Negative-mode spectra, biological samples and instrument-shifted data would establish a broader domain of validity. These extensions should retain the same reporting contract: predictive fit, per-spectrum sparsity, global inventory use, MAG-retrieved and SOS-evaluable motif breadth, and conditional chemical quality.

## 6 Limitations

This study uses one positive-ion-mode MS<sup>n</sup>Lib benchmark and one fixed train/validation/test partition. The three training seeds change initialization and minibatch order but not the data split, vocabulary, topic count or hyperparameters. Their sample standard deviations are descriptive and should not be interpreted as confidence intervals over future datasets or alternative splits. The held-out test results measure final performance on this fixed benchmark, not performance under an external library or instrument shift.

The synthetic simulator supplies known truth but cannot represent the complete chemical, instrumental and noise structure of measured spectra. Its recovery metrics establish mechanism behavior under controlled conditions rather than biological validity.

The simplification review is subsequent validation development, not a second sealed test study. It uses fixed, published-inspired recipes rather than exhaustive per-family tuning, and changes encoder width and decoder capacity between families. The initial ETM ablations use a matched recipe; the expanded family comparisons do not isolate every training choice. Most candidates have only one real seed; three selected alternatives have two additional paired repeats, including near misses under the old numerical flags. PoE topic prototypes also have a different probabilistic meaning from additive emissions. Consequently these comparisons support the full-document-context development recommendation, not proof of equivalence, statistical superiority or global minimality. The 100-unit MLP and fixed-context-weight forms remain plausible further simplifications with insufficient repeat evidence. Future selection needs a new untouched split or external dataset; the earlier test results cannot serve as unbiased confirmation of subsequent decisions.

The observation model treats rounded relative intensities as multinomial pseudo-counts. Those copies are weights, not independent ion observations, and their absolute scaling changes the balance between reconstruction and KL divergence. The fixed 50/50 fragment/loss mass is an explicit domain prior that removes vocabulary-size bias but may not be optimal for every acquisition regime.

MAG and SOS depend on the Spec2Vec reference library, leakage filter, retrieval and clustering settings, MACCS fingerprint bits, 80% consensus threshold and dominant-topic molecule-association rule. The rule gives every model the same number of spectrum associations but represents only the strongest topic for a spectrum; it does not score secondary mixture memberships. SOS is an asymmetric containment measure and does not establish an exact substructure. Reference library coverage can also influence which topics become evaluable.

Finally, the runtime measurements describe one implementation and GPU system. The dense ETM encoder scales with vocabulary size, and peak memory or throughput will change with hardware, batch size, numerical precision and software versions.

## 7 Data, code and reproducibility

The benchmark input and positive-ion-mode annotation assets are available from Zenodo record 20179680 [27]. The repository records the exact data preparation protocol, model implementation, compact result packages, aggregate chemical scores, epoch-level training records, environment manifest and SHA-256 file checksums. The compact evidence bundle includes separate validation and final test results together with manifests that record stage order, fitted checkpoint identity and file provenance.

The historical result tables and macros are generated from frozen machine-readable evidence files in JavaScript Object Notation (JSON) and comma-separated values (CSV) formats:

```
python -m scripts.run_contextual_sparse_etm_reproduction initialize --root RUN
python -m scripts.run_contextual_sparse_etm_reproduction run --root RUN
python -m scripts.package_contextual_sparse_etm_reproduction --root RUN --output EVIDENCE
python -m scripts.generate_contextual_sparse_etm_report --evidence-root EVIDENCE
```

The first two commands initialize and execute the complete acquisition-to-test stage plan. The third verifies stage chronology, file ownership, split release, probability matrices and MAG exception counts before creating an atomic compact package. The final command rechecks the package seal and regenerates the LaTeX tables. The independent review has a separate validation-only inventory, including negative results, input/source hashes and the explicitly staged follow-on hypothesis, under `research/minimal_neural_etm/review_20260907/`. Its tables are regenerated with:

```
python -m scripts.generate_minimal_etm_review
python -m scripts.generate_published_neural_review
python -m scripts.generate_contextual_reduction_review
```

None of these commands modifies the historical evidence seal. New tests independently check Gaussian and Dirichlet KL calculations, entmax-only changes, frozen normalization state and additive versus PoE completion probabilities. The prototypes add no Pyro dependency to production; they use the existing PyTorch environment and remain excluded from the production wheel. The MLP-free input boundary is implemented in `benchmarks/neural_ms2lda/attention_etm.py` and tested against the unchanged ablation implementation. The validation-only compatibility audit is reproducible with `scripts.audit_attention_etm_interface`, using `-run` for an evidence-only validation run and `-output` for its JSON audit. No new fit or foundation-model download is required.

The focused equation-code test in `benchmarks/neural_ms2lda/tests/test_contextual_sparse_etm.py` independently reconstructs the channel-balanced  $\beta$ , contextual top-2 evidence, centered posterior mean, analytic KL divergence, threshold form of 1.5-entmax, raw-count reconstruction and learned-parameter count, then compares each quantity with the implementation.

## 8 Conclusion

Contextual Sparse ETM adapts ETM’s embedding decoder and latent-space ELBO by adding equal fragment/loss decoder mass, contextual top-2 posterior evidence and a 1.5-entmax latent-to-simplex link. Only one learned scalar is added; the induced topic-mixture prior differs from ETM’s logistic-normal prior.

On the compound- and scaffold-disjoint  $MS^n$ Lib test split, the model yields more SOS-evaluable and useful motifs than canonical ETM, fragment/loss-balanced ETM and Tomotopy. Its gains over Tomotopy are 20.8% and 16.2% for the representative initialization, while sparse spectrum assignments coexist with 917 test-set topic winners. All three initializations retain a breadth advantage, with 415–432 useful motifs. Tomotopy retains higher conditional mean SOS and useful/evaluable rate, and dense ETM retains better completion likelihood. The evidence supports increased chemically assessable motif breadth on this split, with quantified quality and predictive trade-offs. Additional datasets are needed to assess generalization. The subsequent validation-only review supports a simpler development form: retain the standard ETM MLP but replace leave-one-out context with the whole-spectrum mean. Three paired seeds preserve useful-motif breadth and compact mixtures with slightly better predictive fit. A shallow 100-unit MLP is a promising, much smaller one-seed alternative; its confirmation remains open. Entmax has direct support for compactness and channel balancing for chemical breadth, while the necessity of every retained attention setting is not proved. The newly recommended form has not received an unbiased held-out test evaluation: the historical test scores above remain attached to the original model. This is a defensible, bounded simplification decision, not a claim that the absolute minimum neural model has been found.

## A Supporting simplification evidence

The tables below retain both favorable and unfavorable development results. Synthetic cells are means over seeds 11, 23 and 37; sample standard deviations and every per-seed result are retained in the machine-readable review summary. They are descriptive repeated-seed comparisons, not confidence intervals over new datasets. An effective-topic count is  $\exp(H(\theta_d))$ , summarized by the median across spectra and then averaged across these three runs.

| $K$ | Model                          | $\beta$ rec. | $\theta$ rec. | Recovered | NLL   | Eff. topics | Near- $\beta$ |
|-----|--------------------------------|--------------|---------------|-----------|-------|-------------|---------------|
| 36  | Contextual (paired)            | 0.465        | 0.754         | 8.00      | 6.287 | 2.00        | 0.746         |
| 36  | BN-entmax, learned scale       | 0.568        | 0.858         | 12.67     | 6.181 | 1.40        | 0.762         |
| 36  | BN-entmax, fixed scale         | 0.596        | 0.885         | 13.67     | 6.146 | 1.52        | 0.735         |
| 36  | Contextual, no channel balance | 0.467        | 0.741         | 8.33      | 6.287 | 1.98        | 0.734         |
| 128 | Contextual (paired)            | 0.641        | 0.910         | 16.67     | 6.188 | 1.67        | 0.885         |
| 128 | BN-entmax, learned scale       | 0.641        | 0.914         | 15.67     | 6.122 | 1.34        | 0.920         |
| 128 | BN-entmax, fixed scale         | 0.666        | 0.939         | 15.33     | 6.095 | 1.51        | 0.924         |
| 128 | Contextual, no channel balance | 0.639        | 0.917         | 16.33     | 6.185 | 1.74        | 0.884         |

Table 10: Initial paired synthetic simplification screen: matched beta and theta cosine recovery, recovered motifs out of 18 at beta cosine at least 0.50, completion negative log-likelihood (NLL), median effective-topic count, and mean nearest-topic beta cosine. Every run is finite and avoids the strict catastrophic duplicate definition; a high near-beta value can still indicate redundant surplus topics. These results do not establish real chemical quality.

| Model                     | Evaluable | Useful | Mean SOS | NLL   | Eff. topics | Winners |
|---------------------------|-----------|--------|----------|-------|-------------|---------|
| BN, learned scale         | 201       | 123    | 0.627    | 8.757 | 39.56       | 288     |
| BN + prior, learned scale | 47        | 26     | 0.626    | 9.025 | 2.96        | 56      |
| BN, fixed scale           | 459       | 296    | 0.651    | 8.231 | 635.33      | 568     |
| BN + prior, fixed scale   | 234       | 156    | 0.656    | 8.781 | 249.25      | 256     |

Table 11: Audited prior-study real validation, after normalization recalibration on training rows only. “Prior” denotes the Gaussian Laplace approximation to a symmetric Dirichlet with per-topic concentration 0.02, not an exact Dirichlet. Column definitions follow Table 7. Compactness, chemistry and inventory breadth do not improve jointly.

| $K$ | Model                          | $\beta$ rec. | $\theta$ rec. | Recovered | NLL   | Eff. topics | Near- $\beta$ |
|-----|--------------------------------|--------------|---------------|-----------|-------|-------------|---------------|
| 36  | ProdLDA (Pyro-reference port)  | 0.716        | 0.848         | 16.00     | 6.793 | 15.49       | 0.283         |
| 36  | Direct Dirichlet, PoE          | 0.503        | 0.575         | 10.33     | 6.551 | 31.07       | 0.166         |
| 36  | Direct Dirichlet, additive LDA | 0.236        | 0.661         | 0.00      | 7.112 | 30.53       | 0.991         |
| 36  | Direct Dirichlet, additive ETM | 0.241        | 0.572         | 0.00      | 7.066 | 29.92       | 0.988         |
| 36  | ProdLDA + entmax               | 0.804        | 0.933         | 17.67     | 6.299 | 1.96        | 0.537         |
| 128 | ProdLDA (Pyro-reference port)  | 0.771        | 0.898         | 18.00     | 6.825 | 59.46       | 0.634         |
| 128 | Direct Dirichlet, PoE          | 0.322        | 0.576         | 1.67      | 6.513 | 111.45      | 0.193         |
| 128 | Direct Dirichlet, additive LDA | 0.244        | 0.725         | 0.00      | 7.114 | 104.75      | 0.995         |
| 128 | Direct Dirichlet, additive ETM | 0.254        | 0.655         | 0.00      | 7.063 | 103.88      | 0.992         |
| 128 | ProdLDA + entmax               | 0.834        | 0.923         | 18.00     | 6.352 | 4.34        | 0.815         |

Table 12: Expanded synthetic screen, with column meanings as in Table 10. PoE denotes product of experts; its beta recovery concerns one-hot conditional prototypes, not identifiable additive emissions. The generator supplies additive truth, so these are useful motif diagnostics rather than a family-neutral likelihood ranking. The entmax row is a separately recorded follow-on after the four stock/adapted recipes.

| $K$ | Reduction                             | $\beta$ rec. | $\theta$ rec. | Recovered | NLL   | Eff. topics | Old flag |
|-----|---------------------------------------|--------------|---------------|-----------|-------|-------------|----------|
| 128 | No context, $r = xA$                  | 0.547        | 0.803         | 13.00     | 6.251 | 1.62        | Fail     |
| 128 | Full-document context                 | 0.630        | 0.915         | 15.67     | 6.193 | 1.72        | Pass     |
| 128 | Fixed context weight $c = 1$          | 0.636        | 0.915         | 16.33     | 6.173 | 1.69        | Pass     |
| 128 | All-topic routing                     | 0.304        | 0.478         | 3.33      | 6.507 | 1.88        | Fail     |
| 128 | Linear evidence offset                | 0.657        | 0.844         | 16.00     | 6.502 | 1.01        | Pass     |
| 128 | Softmax instead of entmax             | 0.537        | 0.809         | 11.00     | 6.195 | 3.33        | Fail     |
| 128 | One hidden layer                      | 0.633        | 0.922         | 15.00     | 6.174 | 1.59        | Pass     |
| 128 | Evidence-only encoder                 | 0.660        | 0.930         | 16.33     | 6.141 | 2.01        | Pass     |
| 128 | Document context + one layer          | 0.627        | 0.905         | 15.00     | 6.180 | 1.59        | Pass     |
| 128 | Document context + fixed $c$          | 0.622        | 0.911         | 15.00     | 6.184 | 1.74        | Pass     |
| 128 | Document + fixed $c$ + one layer      | 0.607        | 0.878         | 14.33     | 6.199 | 1.65        | Fail     |
| 128 | Scaled all-topic routing              | 0.424        | 0.690         | 8.67      | 6.328 | 1.70        | Fail     |
| 128 | Fixed $c$ + hard top-1                | 0.654        | 0.936         | 15.67     | 6.206 | 1.46        | Pass     |
| 128 | One layer, width 100                  | 0.658        | 0.935         | 17.00     | 6.142 | 1.67        | Pass     |
| 128 | Attention-only + document + fixed $c$ | 0.656        | 0.922         | 16.00     | 6.104 | 2.76        | Pass     |
| 36  | No context, $r = xA$                  | 0.406        | 0.619         | 7.33      | 6.362 | 2.28        | Fail     |
| 36  | Full-document context                 | 0.459        | 0.724         | 9.00      | 6.279 | 2.05        | Pass     |
| 36  | Fixed context weight $c = 1$          | 0.455        | 0.739         | 8.67      | 6.285 | 2.01        | Pass     |
| 36  | Linear evidence offset                | 0.586        | 0.846         | 13.67     | 6.309 | 1.54        | Pass     |
| 36  | Softmax instead of entmax             | 0.443        | 0.699         | 8.67      | 6.246 | 3.27        | Pass     |
| 36  | One hidden layer                      | 0.470        | 0.738         | 9.33      | 6.272 | 1.97        | Pass     |
| 36  | Evidence-only encoder                 | 0.595        | 0.843         | 13.33     | 6.216 | 2.00        | Pass     |
| 36  | Document context + one layer          | 0.447        | 0.682         | 9.33      | 6.276 | 2.02        | Pass     |
| 36  | Document context + fixed $c$          | 0.452        | 0.732         | 8.00      | 6.279 | 2.03        | Pass     |
| 36  | Document + fixed $c$ + one layer      | 0.444        | 0.682         | 9.00      | 6.277 | 2.08        | Pass     |
| 36  | Scaled all-topic routing              | 0.362        | 0.562         | 5.33      | 6.395 | 2.10        | Fail     |
| 36  | Fixed $c$ + hard top-1                | 0.494        | 0.745         | 9.33      | 6.296 | 1.95        | Pass     |
| 36  | One layer, width 100                  | 0.490        | 0.758         | 10.67     | 6.255 | 2.02        | Pass     |
| 36  | Attention-only + document + fixed $c$ | 0.556        | 0.804         | 13.00     | 6.188 | 2.76        | Pass     |

Table 13: Complete within-model synthetic screen, means of three seeds per row. Recovery and NLL definitions follow Table 10; “Old flag” preserves the original staged numerical reference, not the final scientific recommendation. The paired current  $K = 128$  reference has beta recovery 0.641, theta recovery 0.910, 16.67 recovered motifs, NLL 6.188 and 1.67 effective topics. All-topic routing was not scheduled at  $K = 36$  after its high- $K$  failure; the separate scaled control was tested at both sizes. The table contains every completed group, including adaptive near-miss follow-ups. Sample SDs and individual runs remain in the source inventory.



| Model, seed                                | Evaluable | Useful                           | Mean SOS | NLL   | Eff. topics | Winners |
|--------------------------------------------|-----------|----------------------------------|----------|-------|-------------|---------|
| Current, seed 42                           | 675       | 394                              | 0.623    | 9.531 | 3.72        | 829     |
| Attention-only + document + fixed $c$ , 42 | 517       | 333                              | 0.643    | 9.057 | 4.00        | 605     |
| Full-document context, 42                  | 669       | 389                              | 0.628    | 9.528 | 3.69        | 806     |
| Document context + fixed $c$ , 42          | 674       | 379                              | 0.620    | 9.452 | 3.68        | 817     |
| Document + fixed $c$ + one layer, 42       | 621       | 363                              | 0.627    | 9.641 | 3.52        | 769     |
| Document context + one layer, 42           | 646       | 353                              | 0.618    | 9.756 | 3.54        | 791     |
| Evidence-only encoder, 42                  | 537       | 326                              | 0.631    | 9.123 | 4.37        | 631     |
| Fixed context weight $c = 1$ , 42          | 643       | 374                              | 0.628    | 9.388 | 3.54        | 796     |
| Fixed $c$ + hard top-1, 42                 | 613       | 379                              | 0.635    | 9.364 | 2.81        | 773     |
| Linear evidence offset, 42                 |           | Training failed: non-finite loss |          |       |             |         |
| No context, $r = xA$ , 42                  | 617       | 366                              | 0.630    | 9.575 | 3.73        | 771     |
| One hidden layer, 42                       | 672       | 402                              | 0.625    | 9.764 | 3.60        | 830     |
| One layer, width 100, 42                   | 625       | 377                              | 0.631    | 9.680 | 3.65        | 790     |
| Softmax instead of entmax, 42              | 662       | 395                              | 0.630    | 9.125 | 47.58       | 759     |
| Current, seed 11                           | 662       | 405                              | 0.629    | 9.537 | 3.77        | 827     |
| Full-document context, 11                  | 671       | 390                              | 0.628    | 9.500 | 3.75        | 817     |
| Evidence-only encoder, 11                  | 530       | 327                              | 0.645    | 9.104 | 4.40        | 623     |
| One hidden layer, 11                       | 677       | 394                              | 0.624    | 9.769 | 3.62        | 819     |
| Current, seed 23                           | 662       | 370                              | 0.622    | 9.552 | 3.79        | 825     |
| Full-document context, 23                  | 657       | 393                              | 0.632    | 9.545 | 3.73        | 799     |
| Evidence-only encoder, 23                  | 571       | 371                              | 0.648    | 9.120 | 4.36        | 660     |
| One hidden layer, 23                       | 657       | 406                              | 0.634    | 9.782 | 3.60        | 809     |

Table 14: All within-model real-validation outcomes,  $K = 1000$ . Definitions follow Table 7; displayed seeds have 7001 added for training. The current seed-42 row is reused from phase 1; 20 new rows are complete and the linear-offset row records a numerical training failure, not zero chemical performance. One-layer means width 800 except the explicitly named width-100 run. No result is a new test estimate.

| Model, seed                                | SOS $\geq 0.5$          | $\geq 0.6$ | $\geq 0.7$ | $\geq 0.8$ | $\geq 0.9$ |
|--------------------------------------------|-------------------------|------------|------------|------------|------------|
| Current, seed 42                           | 562                     | 394        | 192        | 61         | 19         |
| Attention-only + document + fixed $c$ , 42 | 438                     | 333        | 181        | 69         | 23         |
| Full-document context, 42                  | 557                     | 389        | 197        | 75         | 18         |
| Document context + fixed $c$ , 42          | 555                     | 379        | 187        | 71         | 12         |
| Document + fixed $c$ + one layer, 42       | 507                     | 363        | 199        | 72         | 15         |
| Document context + one layer, 42           | 531                     | 353        | 172        | 60         | 14         |
| Evidence-only encoder, 42                  | 455                     | 326        | 168        | 66         | 17         |
| Fixed context weight $c = 1$ , 42          | 547                     | 374        | 195        | 55         | 17         |
| Fixed $c$ + hard top-1, 42                 | 519                     | 379        | 193        | 75         | 25         |
| Linear evidence offset, 42                 | No completed evaluation |            |            |            |            |
| No context, $r = xA$ , 42                  | 511                     | 366        | 183        | 78         | 20         |
| One hidden layer, 42                       | 554                     | 402        | 195        | 62         | 14         |
| One layer, width 100, 42                   | 533                     | 377        | 188        | 67         | 15         |
| Softmax instead of entmax, 42              | 536                     | 395        | 209        | 92         | 32         |
| Current, seed 11                           | 558                     | 405        | 193        | 68         | 17         |
| Full-document context, 11                  | 566                     | 390        | 201        | 72         | 20         |
| Evidence-only encoder, 11                  | 448                     | 327        | 196        | 79         | 23         |
| One hidden layer, 11                       | 577                     | 394        | 189        | 61         | 13         |
| Current, seed 23                           | 552                     | 370        | 184        | 62         | 18         |
| Full-document context, 23                  | 545                     | 393        | 217        | 80         | 22         |
| Evidence-only encoder, 23                  | 497                     | 371        | 198        | 78         | 28         |
| One hidden layer, 23                       | 560                     | 406        | 199        | 69         | 15         |

Table 15: Chemical-cutoff sensitivity on the same validation annotations. Cells count evaluable motifs at inclusive SOS cutoffs; the 0.6 column is the established useful count. The 0.8 column is inclusive, unlike the historical “high” band defined as strictly greater than 0.8. These correlated summaries are descriptive checks, not extra experimental replications or a redefinition of usefulness. The failed linear-offset run has no evaluation.

## B Exact hyperparameters

Table 16: Frozen settings for the reported model and evaluation.

| Quantity                                          | Value                                                                    |
|---------------------------------------------------|--------------------------------------------------------------------------|
| Data split seed                                   | 42                                                                       |
| Training seeds                                    | 7043, 23, 37                                                             |
| Topics                                            | 1000                                                                     |
| Vocabulary                                        | 21,233 train-only fragment/loss tokens                                   |
| Token coordinates                                 | 48-dimensional train-only skip-gram with negative sampling (SGNS); fixed |
| Encoder                                           | 21,233 $\rightarrow$ 800 $\rightarrow$ 800; rectified linear unit (ReLU) |
| Variational outputs                               | 1,000-dimensional $\mu$ and $\log \sigma^2$                              |
| Topic-word decoder                                | Embedded Topic Model (ETM) inner products; 50/50 fragment/loss mass      |
| Contextual evidence                               | leave-one-out context; top-2; temperature 1.0                            |
| Evidence pseudocount                              | fixed $1/K$                                                              |
| Numerical normalization floor                     | $10^{-12}$                                                               |
| Additional learned parameters                     | 1 context scalar                                                         |
| Document-topic probability transform              | 1.5-entmax                                                               |
| Reconstruction                                    | raw pseudo-count multinomial negative log-likelihood                     |
| Prior and KL divergence                           | standard-normal analytic Gaussian Kullback–Leibler divergence            |
| Optimizer                                         | Adam                                                                     |
| Learning rate; weight decay                       | 0.005; $1.2 \times 10^{-6}$                                              |
| Batch size; epochs                                | 256; 120                                                                 |
| Device; host worker threads                       | CUDA; 6                                                                  |
| Mass2Motif Annotation Guidance (MAG) query        | top 20 words; search 1000; 10 unique molecules                           |
| MAG and substructure overlap score (SOS) settings | one dominant topic per spectrum; cluster cosine 0.9; consensus 0.8       |

## C Equation–implementation correspondence

Table 17: Map from the mathematical description to executable repository code.

| Report component                                | Implementation                                                                                                  |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Channel-balanced $\beta$ (Eq. 2)                | <code>contextual_sparse_etm.py:</code><br><code>channel_balanced_topic_word_distribution</code>                 |
| Leave-one-out context (Eq. 7)                   | <code>contextual_sparse_etm.py: leave_one_out_context</code>                                                    |
| Contextual top-2 evidence (Eqs. 8–11)           | <code>contextual_sparse_etm.py: contextual_top2_evidence</code>                                                 |
| Posterior offset (Eq. 12)                       | <code>contextual_sparse_etm.py: centered_log_evidence_offset</code>                                             |
| Gaussian posterior and KL (Eqs. 6, 19)          | <code>contextual_sparse_etm.py: posterior; diagonal_gaussian_kl</code>                                          |
| 1.5-entmax $\theta$ (Eqs. 14, 15)               | <code>contextual_sparse_etm.py: entmax15_document_mixture</code>                                                |
| Reconstruction loss (Eq. 18)                    | <code>topic_model_training.py: raw_count_reconstruction_loss</code>                                             |
| Real training and deterministic inference       | <code>scripts/run_contextual_sparse_etm.py</code>                                                               |
| Synthetic preparation and ablations             | <code>prepare_contextual_sparse_etm_synthetic.py;</code><br><code>run_contextual_sparse_etm_synthetic.py</code> |
| Published ETM controls                          | <code>etm_baselines.py; run_etm_controls.py</code>                                                              |
| Document completion                             | <code>model_evaluation.py: completion_metrics</code>                                                            |
| Dominant-topic association and SOS (Eqs. 24–25) | <code>chemical.py: _associated_record_indices; _topic_scores</code>                                             |
| Equation–code correspondence test               | <code>tests/test_contextual_sparse_etm.py</code>                                                                |
| Clean-room orchestration                        | <code>study_protocol.py; reproduction_plan.py;</code><br><code>run_contextual_sparse_etm_reproduction.py</code> |
| Evidence verification and packaging             | <code>reproduction_audit.py; evidence_bundle.py; report_evidence.py</code>                                      |

## D Metric interpretation

| Metric                                            | Definition                                                                                                                                  | Reading                                       |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Completion negative log-likelihood (NLL)          | Count-weighted log loss on withheld in-vocabulary words after observing the other physical peak groups.                                     | Lower is better predictive fit.               |
| Optimized motifs                                  | Topics for which Mass2Motif Annotation Guidance (MAG) retains at least one fragment or neutral loss.                                        | Breadth of annotatable topic spectra.         |
| Substructure-overlap-score (SOS)-evaluable motifs | Optimized topics with at least one valid unique test molecule whose spectrum has that topic as its largest full-spectrum mixture component. | Inventory eligible for chemical scoring.      |
| Mean/median SOS                                   | Containment of the MAG consensus fingerprint in associated molecules, averaged first within motif and then across motifs.                   | Higher is better conditional consistency.     |
| Useful motifs                                     | SOS-evaluable motifs with SOS at least 0.6.                                                                                                 | Absolute count in intermediate or high bands. |
| Effective topics                                  | Exponentiated entropy of one spectrum’s topic mixture.                                                                                      | Lower means a more concentrated mixture.      |
| Unique top-1 topics                               | Topics that are the largest mixture component for at least one test spectrum.                                                               | Breadth of document-level topic use.          |

Table 18: Interpretation of the main predictive, chemical and inventory measures. Counts and conditional averages answer different questions.

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